

Fractional Quantum Geometry in Topological Mott Regime

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Inspired by recent interests in quantum geometry in electronic bands, we consider a two-dimensional topological Mott insulator, *i.e.* chiral spin liquid, whose spinon bands develop nontrivial band topological and quantum geometry. The virtual polarization of the gapped charge sector transfers the electromagnetic drive to the charge-neutral spinons through the emergent $U(1)$ gauge field, producing a tensor Ioffe-Larkin response and enabling two complementary optical protocols for measurements. In the off-shell (low-frequency) regime, a self-calibrated ratio of the physical longitudinal and Hall conductivities yields the spinon Chern number without the microscopic knowledge of the charge response. In the on-shell (resonant) regime, combining the inverse Ioffe-Larkin response with the Kramers-Kronig relation enables reconstruction of the quantum-metric and Berry-curvature spectral densities. Our approach directly applies to the triangular lattice Hofstadter-Hubbard model in Moiré systems, can be well adapted to other triangular lattice antiferromagnets such as Nb_3Br_8 with appropriate modifications. We establish a unified framework for quantitatively probing the quantum geometry of fractionalized quasiparticles in topological Mott regime.

Introduction.—The quantum geometric tensor plays a pivotal role in quantum materials, governing transport, optical, and collective phenomena beyond what is encoded in the band dispersion [1, 2]. Its antisymmetric part, the Berry curvature, determines Hall-type responses, while its symmetric part, the quantum metric, controls interband spectral weight, Wannier localization, and geometric contributions to flat-band superfluidity [3–7]. Electromagnetic probes can access these quantities for charged quasiparticles [8, 9], but not directly for the fractionalized, electrically neutral excitations of a quantum spin liquid in the Mott regime [10–13].

This obstruction is especially interesting in the topological Mott regime, where the neutral excitations may develop the nontrivial band structure topology even though the charge transport is frozen [14–18]. One representative example in 2D is the chiral spin liquid (CSL) where the spinon band has a nontrivial Chern number. Recently, several materials, including the moiré heterostructure [19, 20], the triangular-lattice organics [17, 21], and the cluster magnets Nb_3Br_8 [22] have been proposed to realize the CSL. Unlike an ordinary Chern insulator, a CSL has no electronic dc Hall conductance from which the spinon Chern number can be read off directly. Previous studies have investigated the sub-Mott-gap optical and magneto-optical signatures of spinon Fermi surfaces and fractionalization [23–27], as well as electromagnetic signatures of chiral order [28]. More recently, Raman circular dichroism has been shown to be sensitive to the quantum geometry and handedness of fractionalized excitations [29]. These works primarily identified qualitative spectroscopic signatures of fractionalization and chirality. Here, we instead develop a quantitative optical protocol that explicitly extracts the spinon Chern number and reconstructs the hidden quantum geometry of gapped spinon bands without a Fermi surface.

We start with a slave-rotor formulation that simultaneously captures the charge and spin fluctuations together in a CSL. In this framework, a sub-Mott-gap electromagnetic field can virtually polarize the gapped charge sector, thereby

driving neutral spinons through the induced internal gauge field [23, 30], as illustrated in Fig. 1. This mediation produces a frequency-dependent Mott filter that dresses the spinon response. Consequently, the physical optical response is not a direct measure of the spinon quantum geometry, but rather a tensor Ioffe-Larkin composition of the chargon and spinon responses. In this Letter, we show that the off-shell, low-frequency longitudinal and Hall responses together determine the spinon Chern number without independent knowledge of the charge polarizability, whereas the on-shell resonant responses reconstruct the energy-resolved quantum-metric and Berry-curvature spectral densities.

Parton optical response.—We decompose the electron operator as $c_{i\sigma} = b_i f_{i\sigma}$, where $b_i = e^{i\theta_i}$ is the charged ro-

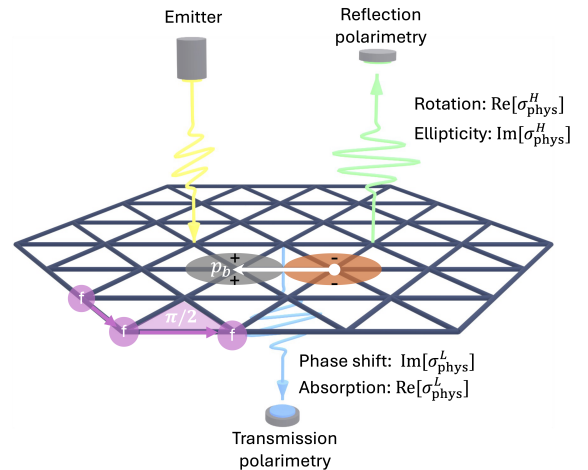


FIG. 1. Optical probe of spinon quantum geometry on a triangular lattice CSL. Linearly polarized incident light (yellow) couples to the charge sector and, through imbalanced doublon-holon fluctuations, induces a virtual polarization p_b (white arrow), which mediates the coupling to the internal gauge field seen by the spinons (purple). Transmission and reflection polarimetry resolve the longitudinal and Hall conductivities, respectively.

tor (chargon) and $f_{i\sigma}$ is a neutral spinon [31, 32]. As a specific model, we consider the half-filled triangular-lattice Hofstadter–Hubbard (TLHH) model with electron hopping t , repulsion U , and orbital flux $\Phi_\Delta = \pm\pi/2$ per triangle plaquette from the external magnetic field. In the uniform CSL solution, the spinons entirely inherit this flux while the chargons experience zero static flux, thus becoming trivially gapped [33–35]. Writing the static electromagnetic and internal backgrounds as $\bar{\mathbf{A}}$ and $\bar{\mathbf{a}}$, respectively, this saddle point admits the gauge choice $\bar{\mathbf{a}} = \bar{\mathbf{A}}$. We use the unbarred fields $\mathbf{A}$ and $\mathbf{a}$ to denote the optical probe and the internal gauge fluctuation about these backgrounds. Therefore, the spinons couple to $\mathbf{a}$, whereas the chargons couple to $\mathbf{A} - \mathbf{a}$. Since $\bar{\mathbf{a}} = \bar{\mathbf{A}}$, the chargon sector indeed experiences zero average flux. Throughout this Letter, we use the natural units $e = \hbar = k_B = 1$ for simplicity.

Integrating out the parton fields yields the effective quadratic action, which in the uniform optical limit has the form [36],

$$S^{\text{eff}}[A, a] = \frac{1}{2}(A - a)_i \Pi_b^{ij} (A - a)_j + \frac{1}{2} a_i \Pi_f^{ij} a_j. \quad (1)$$

Here, Π_b and Π_f are the chargon and spinon response kernels that depend on the driving frequency ω , with $\{i, j, k\} \in \{x, y, z\}$. Both spatial and frequency sums are assumed implicitly. Integrating out the gauge field $\mathbf{a}$ yields the tensor Ioffe–Larkin rule for the physical response kernel [30, 37, 38],

$$\Pi_{\text{phys}} = \Pi_b - \Pi_b(\Pi_b + \Pi_f)^{-1} \Pi_b. \quad (2)$$

where we have omitted the spatial indices, and the inverse relation reads $\Pi_{\text{phys}}^{-1} = \Pi_b^{-1} + \Pi_f^{-1}$. The physical optical conductivity is given by $\sigma_{\text{phys}}(\omega) = i\Pi_{\text{phys}}(\omega)/\omega$.

In the CSL Mott regime, the chargon is gapped and thus is not condensed. The chargon has a real hopping and a vanishing static relative flux. Its antiunitary symmetry forbids a Hall response at all frequencies, while the unbroken C_3 rotation symmetry makes the longitudinal response isotropic [36]. Thus, we have the decomposition along the longitudinal and Hall channels for the spinon and chargon kernels as

$$\Pi_b^{ij}(\omega) = p_b(\omega)\delta^{ij}, \quad \Pi_f^{ij}(\omega) = \Pi_f^L(\omega)\delta^{ij} + \Pi_f^H(\omega)\varepsilon^{ij}, \quad (3)$$

with ε^{ij} the Levi-Civita symbol. Below the chargon pair energy threshold $\omega_b^{\text{th}} = 2\Delta_b$, the real doublon-holon pair creation is forbidden; virtual doublon-holon fluctuations instead produce an electric polarization response. The clean chargon kernel is therefore real, and we denote this scalar kernel by p_b to emphasize its polarization character. Its low-frequency expansion is [36]

$$p_b(\omega) = -\chi_b\omega^2 + O(\omega^4). \quad (4)$$

where $\chi_b = -\lim_{\omega \rightarrow 0} p_b/\omega^2$ is the chargon polarizability and the minus sign is consistent with a positive electric polarizability convention. The full expression for the Gaussian char-

gon response is derived in the SM [36]. For an isotropic tensor $X^{ij} = X^L\delta^{ij} + X^H\varepsilon^{ij}$, define the circular eigenvalues $X^\pm = X^L \pm iX^H$. Eq. (2) then becomes

$$\Pi_{\text{phys}}^\pm = \frac{p_b \Pi_f^\pm}{p_b + \Pi_f^\pm} \equiv \mathcal{F}^\pm \Pi_f^\pm, \quad \mathcal{F}^\pm = \frac{p_b}{p_b + \Pi_f^\pm}. \quad (5)$$

The complex, frequency-dependent Mott filter $\mathcal{F}^\pm$ expresses the mediation of the optical drive by virtual chargon polarization. Consequently, the physical response does not directly measure the pure spinon kernel; instead, it encodes both the chargon and spinon responses in a nontrivial way. The TLHH model provides a perfect example of a gapped CSL where the chargon sector that directly couples to the probe electromagnetic field is trivial, while the spinon sector that couples to the internal gauge field is topologically nontrivial. In the following, we first show how the spinon quantum geometry is encoded in the spinon kernel, and then demonstrate how to extract it from the physical response.

The TLHH with the uniform flux $\Phi_\Delta = \pi/2$ or $-\pi/2$ per triangle plaquette has two magnetic sublattices once the gauge is fixed. For each spin flavor its Bloch Hamiltonian can be written as $h_f(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\tau}$, where $\boldsymbol{\tau}$ acts on the two sublattices. For $\Phi_\Delta = \pi/2$, a convenient gauge gives [36]

$$\mathbf{d}(\mathbf{k}) = -2t_f(\cos k_2, -\cos(k_2 - k_1), \cos k_1), \quad (6)$$

with $k_1 = ak_x, k_2 = a(k_x + \sqrt{3}k_y)/2$, and lattice spacing a . The band energies are $E_\pm = \pm d$, where $d = |\mathbf{d}|$. The lower band is filled for each of the $N_\sigma = 2$ spin flavors and has $C_s = +1$ in our convention. Complex conjugation reverses the flux and C_s , giving the opposite chirality of the $U(1)_{\pm 2}$ CSL [14, 16]. The hopping amplitudes for spinon and chargon are determined by mean-field solutions $t_f = t|\langle b_i^\dagger b_j \rangle|$ and $t_b = t|\sum_\sigma \langle f_{i\sigma}^\dagger f_{j\sigma} \rangle|$, together with the Hilbert space constraint fixed λ . The details of self-consistent solutions are given in the SM [36]. The spinon Δ_s and chargon gaps Δ_b are known once t_f and t_b are determined at a specific value of U/t . The spinon interband threshold is then $\Delta_s = 2\min_{\mathbf{k}} d(\mathbf{k}) = 2\sqrt{3}t_f$, and the interband continuum spans $\Delta_s \leq \omega \leq 2\Delta_s$ [39].

Including both spin flavours, the spinon model has four bands. Since the total spin is conserved and the Hamiltonian separates into two identical two-band blocks, each with one filled lower band. The eigenstates factorize as $|u_{\pm, \sigma}(\mathbf{k})\rangle = |u_{\pm}(\mathbf{k})\rangle \otimes |\sigma\rangle$, with the momentum-independent spin states. The two flavours contribute the identical geometric responses, accounted for by the factor $N_\sigma = 2$ below. We therefore define the quantum geometric tensor of the occupied lower band $|u_{-}\rangle$ within a single spin flavour as

$$Q_{ij} = \langle \partial_i u_{-} | (1 - |u_{-}\rangle\langle u_{-}|) | \partial_j u_{-} \rangle = g_{ij} - \frac{i}{2} \Omega_{ij}. \quad (7)$$

whose real and imaginary parts correspond to the quantum metric g_{ij} and Berry curvature Ω_{xy} , respectively [1–3]. After the imaginary-time path integral and Matsubara summation [36], the zero-temperature retarded kernels are related to

the quantum geometric tensor as

$$\Pi_f^L(\omega) = \mathcal{P} \int [d\mathbf{k}] \frac{4d(\mathbf{k})\omega^2}{\omega^2 - [2d(\mathbf{k})]^2} g_{xx}(\mathbf{k}) - i\pi\omega^2 \int [d\mathbf{k}] g_{xx}(\mathbf{k}) \delta[\omega - 2d(\mathbf{k})], \quad (8)$$

$$\Pi_f^H(\omega) = -\frac{\pi\omega^2}{2} \int [d\mathbf{k}] \Omega_{xy}(\mathbf{k}) \delta[\omega - 2d(\mathbf{k})], -i\mathcal{P} \int [d\mathbf{k}] \frac{4\omega d^2(\mathbf{k}) \Omega_{xy}(\mathbf{k})}{\omega^2 - [2d(\mathbf{k})]^2}, \quad (9)$$

where $\int [d\mathbf{k}] \equiv N_\sigma \int_{\text{MBZ}} d^2k / (2\pi)^2$ runs over the magnetic Brillouin zone and $\mathcal{P}$ denotes the principal value. These expressions connect the spinon longitudinal and Hall kernel (in response to internal gauge fields) with the spinon quantum metric and Berry curvature, providing the basis for the two optical extraction protocols below.

Off-shell measurement.—Our first goal is to extract the spinon Chern number from the measurable subgap conductivities. The starting point is the dc spinon Hall conductivity. Taking the low-frequency limit of Eq. (9) gives

$$\sigma_{f,0}^H \equiv \text{Re } \sigma_f^H(\omega \rightarrow 0) = -\lim_{\omega \rightarrow 0} \frac{\text{Im } \Pi_f^H(\omega)}{\omega} = -\int [d\mathbf{k}] \Omega_{xy}(\mathbf{k}) = -\frac{N_\sigma C_s}{2\pi}. \quad (10)$$

where C_s is the Chern number per spin flavour. Unfortunately, this quantized response is to the internal gauge field and cannot be measured directly. To determine how it enters the physical optical response, we decompose the Ioffe-Larkin rule in Eq. (5) into the longitudinal and Hall channels with

$$\Pi_{\text{phys}}^L = p_b - \frac{p_b^2(p_b + \Pi_f^L)}{(p_b + \Pi_f^L)^2 + (\Pi_f^H)^2}, \quad (11)$$

$$\Pi_{\text{phys}}^H = \frac{p_b^2 \Pi_f^H}{(p_b + \Pi_f^L)^2 + (\Pi_f^H)^2}. \quad (12)$$

The physical Hall response thus inherits the spinon Hall kernel, but its magnitude is modulated by the charge sector. We here use the accompanying longitudinal response to eliminate this unknown contribution in the low-frequency limit. For $0 < \omega < \Delta_s \ll \omega_b^{\text{th}} = 2\Delta_b$, both sectors are nonabsorptive. p_b and Π_f^L are real, and Π_f^H is purely imaginary. Eq. (12) can be simplified as

$$\text{Re } \sigma_{\text{phys}}^H(\omega) = \mathcal{F}^H(\omega) \text{Re } \sigma_f^H(\omega), \quad (13)$$

where $\mathcal{F}^H(\omega)$ is a real Hall Mott filter and has the form

$$\mathcal{F}^H(\omega) = \frac{p_b^2(\omega)}{[p_b(\omega) + \Pi_f^L(\omega)]^2 + [\Pi_f^H(\omega)]^2}. \quad (14)$$

To identify the conductivity combination that cancels the

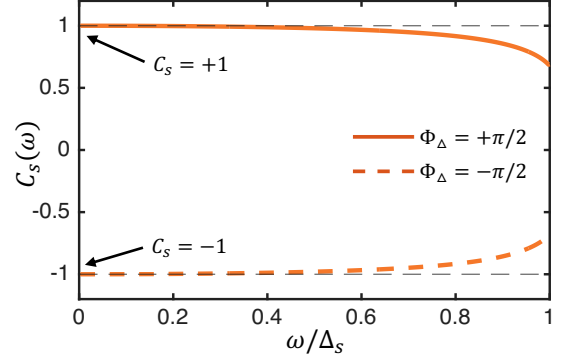


FIG. 2. Finite-frequency Chern-number estimator $C_s(\omega)$ from Eq. (18) for the triangular-lattice CSL at $U/t = 15$. The solid (dashed) orange curves correspond to $\Phi_\Delta = +\pi/2$ ($-\pi/2$), and approach $C_s = +1$ (-1) as $\omega \rightarrow 0$. The horizontal gray dashed lines mark the quantized values.

charge contribution, we expand the kernels at low frequency,

$$p_b = -\chi_b \omega^2 + O(\omega^4), \quad (15a)$$

$$\Pi_f^L = -\chi_f^L \omega^2 + O(\omega^4), \quad (15b)$$

$$\Pi_f^H = -i\omega \sigma_{f,0}^H + O(\omega^3), \quad (15c)$$

where $\chi_f^L = \int [d\mathbf{k}] g_{xx}(\mathbf{k}) / d(\mathbf{k}) > 0$ and $\chi_b > 0$ comes from the definition in Eq. (4). For $C_s \neq 0$, the Hall term controls the leading order of the Ioffe-Larkin denominator. Thus, based on Eq. (14), we have $\mathcal{F}^H = -\chi_b^2 \omega^2 / (\sigma_{f,0}^H)^2 + O(\omega^4)$. On the other hand, substituting the expansions from Eq. (15) into Eq. (11) gives $\Pi_{\text{phys}}^L = -\chi_b \omega^2 + O(\omega^4)$. Collecting all the frequency-dependent terms, we obtain

$$\text{Re } [\sigma_{\text{phys}}^H] = \frac{2\pi \chi_b^2}{N_\sigma C_s} \omega^2 + O(\omega^4), \quad (16)$$

$$\text{Im } [\sigma_{\text{phys}}^L] = -\chi_b \omega + O(\omega^3). \quad (17)$$

From the above expressions, we see that the charge polarizability χ_b appears in both the longitudinal and Hall responses. This leads to a self-calibrated estimator for the spinon Chern number C_s based solely on the measurable physical conductivities, without requiring any microscopic knowledge of the chargon response. Specifically, we define the finite-frequency Chern-number estimator $C_s(\omega)$ as

$$C_s(\omega) \equiv \frac{2\pi}{N_\sigma} \frac{[\text{Im } \sigma_{\text{phys}}^L(\omega)]^2}{\text{Re } \sigma_{\text{phys}}^H(\omega)}. \quad (18)$$

From Eqs. (16) and (17), we know that $C_s(\omega)$ approaches the quantized spinon Chern number $C_s = \lim_{\omega \rightarrow 0} C_s(\omega)$ in the $\omega \rightarrow 0$ limit [40], providing a direct optical probe of the topological invariant of the neutral spinon bands.

To verify our proposal, we numerically evaluate Eq. (18) based on our TLHH model with the self-consistent ansatz. Fig. 2 shows that $C_s(\omega)$ indeed approaches $+1$ and -1 in the

low-frequency limit for the two opposite fluxes. Reversing the flux sign reverses the Hall response and the estimator, while leaving the longitudinal response unchanged. The numerical result also shows that an accurate estimate does not require an asymptotically small probe frequency: even at $\omega/\Delta_s = 0.5$, we have $C_s(\omega) \simeq \pm 0.98$, which is already very close to the ideal quantized value. Therefore, the estimator remains close to the topological invariant over an appreciable subgap frequency window. This is useful experimentally because, in the insulating regime, $\text{Re}[\sigma_{\text{phys}}^H]$ and $\text{Im}[\sigma_{\text{phys}}^L]$ vanish quickly as ω^2 and ω at low frequencies, respectively. Thus, measurement at finite subgap frequencies supports larger measurable signals. Exact quantization refers to extrapolation to the $\omega = 0$ limit. Finally, we note that in the presence of a finite spinon broadening η_s , the estimator deviates from the quantized value for $\omega \lesssim \eta_s$ and diverges as $\omega \rightarrow 0$, although the underlying band Chern number remains unchanged. Reliable extraction therefore requires the window $\eta_s \ll \omega \ll \Delta_s$, as explained by the low-frequency analysis [36].

On-shell measurement.—We next turn to the energy-resolved quantum geometry probed by the spinon interband transitions, for which the on-shell condition is $\omega = 2d(\mathbf{k})$. Define the geometric spectral densities as $O(\omega) = \int [d\mathbf{k}] O(\mathbf{k}) \delta[\omega - 2d(\mathbf{k})]$, where $O = g_{xx}$ or Ω_{xy} . Eqs. (8) and (9) relate these spectra directly to the spinon kernels:

$$g_{xx}(\omega) = -\frac{\text{Im}\Pi_f^L(\omega)}{\pi\omega^2}, \quad \Omega_{xy}(\omega) = -\frac{2\text{Re}\Pi_f^H(\omega)}{\pi\omega^2}. \quad (19)$$

To recover the spinon kernels from the measured physical response, we need to circumvent the unknown charge contribution. In the circular basis, the absence of a chargin Hall response gives us $\Pi_b^\pm = p_b \in \mathbb{R}$. Introducing the inverse responses $R_\pm \equiv 1/\Pi_{\text{phys}}^\pm$, $r_\pm \equiv 1/\Pi_f^\pm$, and $q_b \equiv 1/p_b$, we obtain

$$R_\pm = q_b + r_\pm. \quad (20)$$

The unknown chargin term identically enters in both channels and can be removed by taking their difference. In addition, since we only consider the frequency regime well below the chargin absorption threshold ω_b^{th} , taking the imaginary part of their sum removes the real p_b as well. To express the spinon information isolated by these two combinations, we similarly define the longitudinal and Hall components of the inverse spinon kernel as $r_f^L = (r_+ + r_-)/2$ and $r_f^H = (r_+ - r_-)/(2i)$. They satisfy

$$r_f^H(\omega) = \frac{R_+(\omega) - R_-(\omega)}{2i}, \quad (21a)$$

$$\text{Im} r_f^L(\omega) = \text{Im} \frac{R_+(\omega) + R_-(\omega)}{2}. \quad (21b)$$

These relations determine r_f^H and $\text{Im}[r_f^L]$ directly from optical data. The inverse physical responses are obtained from $R_\pm = i/(\omega\sigma_{\text{phys}}^\pm)$. The four real conductivity components entering $\sigma_{\text{phys}}^\pm$ can be measured in the polarization-resolved

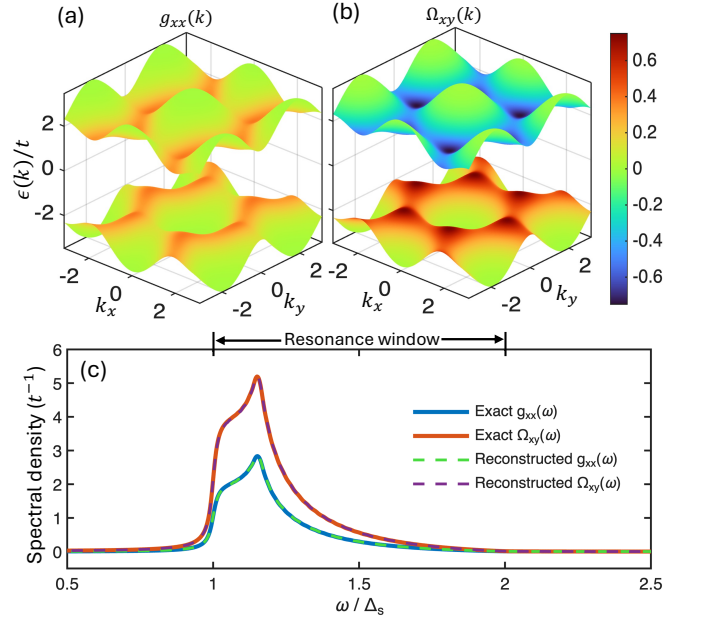


FIG. 3. Spinon quantum geometry and its optical reconstruction for the triangular-lattice CSL. (a,b) Spinon bands colored by the quantum metric $g_{xx}(\mathbf{k})$ and Berry curvature $\Omega_{xy}(\mathbf{k})$, respectively. (c) Geometric spectral densities calculated directly (solid curves) and reconstructed using the inverse-response Kramers–Kronig protocol (dashed curves), at $U/t = 15$ with small spinon broadening $\eta_s/t_f = 0.05$. interband continuum.

reflection and transmission channels [Fig. 1]. Specifically, Kerr rotation and ellipticity in the reflection polarimetry encode $\text{Re}[\sigma_{\text{phys}}^H]$ and $\text{Im}[\sigma_{\text{phys}}^H]$, respectively. In transmission, the phase shift encodes $\text{Im}[\sigma_{\text{phys}}^L]$, whereas $\text{Re}[\sigma_{\text{phys}}^L]$ is inferred from the absorbed intensity after correcting for reflection losses.

The last remaining unknown component, $\text{Re}[r_f^L]$, is determined up to a subtraction constant by the Kramers-Kronig (KK) relation implied by causality:

$$\text{Re}[r_f^L(\omega)] = r_{f,\infty}^L + \frac{2}{\pi} \mathcal{P} \int_0^\infty d\omega' \frac{\omega' \text{Im} r_f^L(\omega')}{\omega'^2 - \omega^2}. \quad (22)$$

Here $r_{f,\infty}^L = \lim_{|z| \rightarrow \infty} r_f^L(z)$ denotes the constant term at infinite frequency since $\Pi_f^L(\infty)$ is finite. The Kramers-Kronig relation requires an analytic inverse kernel. $\Pi_f^\pm(z)$ must not have zeros on the upper half-plane, which is indeed the case in our model. Additionally, the opposite zero-frequency poles of $r_\pm$ cancel in $r_f^L(0) \propto (r_+(0) + r_-(0))$. The analytic conditions and numerical implementation are provided in the SM [36].

For the TLHH CSL ansatz, the constant term $r_{f,\infty}^L$ can be calibrated from the spinon interband threshold. Carrying out the high-frequency expansion of Eq. (8) and Eq. (9) directly gives $r_{f,\infty}^L = (t_f \mathcal{D}_\Delta)^{-1}$, where $\mathcal{D}_\Delta \simeq 1.38681$ is a dimensionless geometrical factor that depends only on the lattice structure and the number of spin flavors [36]. The spinon gap is given by $\Delta_s = 2\sqrt{3}t_f$, which leads to $r_{f,\infty}^L \simeq 2.49788/\Delta_s$. The

interband factors $\delta[\omega - 2d(\mathbf{k})]$ imply that the spinon response has no absorptive weight below $\Delta_s = 2\min_{\mathbf{k}} d(\mathbf{k})$. Thus, the measured leading interband edge ω_f^{th} of $\text{Im}[r_f^L(\omega)]$ identifies $\Delta_s = \omega_f^{th}$ [36]. This determines $r_{f,\infty}^L$ and completes the KK reconstruction. Since $r_f^L(\omega)$ only has a measurable weight in the interband window $[\Delta_s, 2\Delta_s]$, the KK integral can be evaluated with a finite cutoff $2\Delta_s < \Omega_{\text{cut}} < 2\Delta_b$ in the integral (22). We also provide another calibration method for $r_{f,\infty}^L$ based on the first moment of the dynamical spin structure factor [36].

Once r_f^L is completely determined, the spinon kernel follows from

$$\Pi_f^\pm(\omega) = [r_f^L(\omega) \pm ir_f^H(\omega)]^{-1}. \quad (23)$$

Recovering $\Pi_f^L = (\Pi_f^+ + \Pi_f^-)/2$ and $\Pi_f^H = (\Pi_f^+ - \Pi_f^-)/(2i)$ then yields the geometric spectra through Eq. (19). We test this protocol using the self-consistent TLHH ansatz. Figs. 3(a,b) show the spinon bands colored by their quantum metric and Berry curvature. The solid curves in Fig. 3(c) are obtained by directly integrating these microscopic geometric quantities over momentum with the chosen spinon broadening; they serve as the exact benchmark within the mean-field model. For the dashed curves, we first combine the microscopic spinon and chargon kernels through Eq. (2) to generate synthetic optical conductivities. We then treat these conductivities as the measured inputs and apply the inverse-response KK protocol, with the subtraction constant calibrated from Δ_s , to recover the spinon kernels and hence the geometric spectra. The reconstruction protocol itself uses no microscopic chargon kernel as input. Both calculations use the same broadening. The near coincidence of the solid and dashed curves in Fig. 3(c) shows that the protocol accurately reproduces the spectral properties of the spinon quantum geometry, including the peak positions and heights, the relative spectral weights of the metric and curvature, and the broadened tails, thereby validating our inverse-response KK reconstruction theory. In practice, experimental reconstruction requires accurate measurements of the optical response and accurate calibration of Δ_s . The small chargon kernel comparing to the spinon kernel makes the inverse-response extraction sensitive to measurement errors. In principle, the experimental accuracy depends on signal to noise and calibration precision [36].

Discussion.—We have established a quantitative optical framework to extract the spinon Chern number and reconstruct the fractionalized quantum geometry within the topological Mott regime for a CSL. Virtual chargon polarization couples an optical probe to neutral spinons through the emergent gauge field. The resulting tensor Ioffe-Larkin response supports two complementary measurements. A self-calibrated low-frequency conductivity ratio extracts the spinon Chern number, while inverse-response Kramers–Kronig protocol reconstructs the fractionalized quantum geometry. While our numerical calculations utilize the triangular-lattice Hofstadter-Hubbard (TLHH) model, which directly simulates the physics of moiré heterostructures un-

der a strong orbital magnetic flux, the core physical insights and the tensor Ioffe-Larkin inversion protocol developed here hold broader relevance for a wide class of triangular-lattice Mott insulators and other topological Mott insulators.

Potential applications include surfaces of the cluster magnet Nb_3Br_8 [22], 1T-TaS₂ [41–43], and triangular-lattice organic spin liquids such as $\kappa\text{-(BEDT-TTF)}_2\text{Cu}_2(\text{CN})_3$ [21], which have been proposed to host CSLs or related fractionalized phases. For a CSL realized without external magnetic flux $\vec{A} = 0$, however, both the spinons and chargons simultaneously experience a nonzero emergent gauge field. Specifically, the chargons couple to $\vec{A} - \vec{a} = -\vec{a} \neq 0$, which is exactly opposite to the spinon gauge field, which generates a subgap virtual Hall response, $\Pi_b^H \neq 0$. The present extraction protocols must therefore be extended to account for the chargon Hall kernel, whose contribution would require independent constraints from microscopic calculations or additional magneto-optical measurements.

Our established framework also extends beyond two-dimensional CSLs to three-dimensional topological Mott insulators [44] and spinon Weyl semimetals [45] where the nontrivial internal magnetoelectric response may be revealed by physical measurements [46]. Additionally, recent studies of geometric effects in anyon binding and interaction-driven phases of fractionalized quasiparticles [47, 48] motivate a further question, how does spinon quantum geometry influence instabilities [49] driven by residual interactions? In particular, electronic anomalous Hall crystals [50, 51] motivate exploring whether analogous topological crystalline phases can emerge from interacting spinons such that conventional spin orders can coexist with fractionalization [52–55]. These extensions and open questions provide interesting directions for future work.

Acknowledgments.—We thank Jiahao Yang for fruitful discussions. This work was supported by the Quantum Science and Technology–National Science and Technology Major Project (Grant No. 2025ZD0300500), the NSFC (Grant Nos. 92565110 and 12574061), and the BJNSF (Grant No. F261004).

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Supplementary Material for Optical Probes of Spinon Quantum Geometry in Mott-Insulating Chiral Spin Liquids

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SLAVE-ROTOR STRUCTURE OF THE HOFSTADTER–HUBBARD MODEL

We give a microscopic realization of the parton structure used in the main text by starting from the spin-1/2 Hofstadter–Hubbard model on the triangular lattice,

$$H_{\text{HH}} = -t \sum_{\langle ij \rangle, \sigma} \left(e^{i\bar{A}_{ij}} c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.} \right) + \frac{U}{2} \sum_i (n_i - 1)^2. \quad (\text{S1})$$

Here, the site number $n_i = \sum_{\sigma} c_{i\sigma}^\dagger c_{i\sigma}$ accounts for the two (pseudo)spin flavors $\sigma = \{\uparrow, \downarrow\}$, and we work at half filling, $\langle n_i \rangle = 1$, i.e., one electron per site. We specialize to the commensurate Hofstadter flux

$$\sum_{\partial \Delta} \bar{A}_{ij} \equiv \Phi_{\Delta} = \frac{\pi}{2} \pmod{2\pi}$$

through each oriented elementary triangle. This is an externally chosen parameter of the microscopic model, not a quantity determined by the parton saddle. More generally, the flux-matched $SU(N)$ construction takes $\Phi_{\Delta} = \pi/N$. Because a primitive triangular-lattice unit cell contains two elementary triangles, its flux is then $2\pi/N$, so that the magnetic unit cell contains N sites. At one particle per site, a magnetic unit cell containing N sites accommodates N particles. In an $SU(N)$ -symmetric state, this corresponds to one particle per flavor per magnetic unit cell. Since each magnetic subband contains one state per flavor per magnetic unit cell, the lowest $C = 1$ Hofstadter band of each flavor is completely filled. Thus, the total Chern number is $C = N$ for the filled lowest Hofstadter band of the $SU(N)$ model. The present $SU(2)$ problem is the $N = 2$ member of this construction (for $N = 2$, one particle per site corresponds to half filling). The bar on $\bar{A}_{ij}$ distinguishes this fixed background from the weak electromagnetic field used below as an optical probe. Up to an additive constant, the in-

teraction in Eq. (S1) is the usual on-site Hubbard repulsion at half filling.

We then apply the standard slave-rotor decomposition [31]

$$c_{i\sigma} = b_i f_{i\sigma}, \quad (\text{S2})$$

where $b_i = e^{i\theta_i}$ is the charged rotor, referred to as the chargon, and $f_{i\sigma}$ is a neutral spinon carrying spin σ . With $L_i = -i\partial_{\theta_i}$, the physical Hilbert space at half filling is selected by

$$L_i + n_i^f = 1, \quad n_i^f = \sum_{\sigma} f_{i\sigma}^\dagger f_{i\sigma}. \quad (\text{S3})$$

Thus a singly occupied site has $L_i = 0$, whereas an empty (doubly occupied) site has $L_i = +1$ (-1), and the Hubbard term becomes $UL_i^2/2$ in the enlarged Hilbert space. Natural units $e = \hbar = k_B = 1$ are used throughout.

Substitution of Eq. (S2) into the kinetic energy gives the quartic parton hopping

$$H_t = -t \sum_{\langle ij \rangle, \sigma} \left(e^{i\bar{A}_{ij}} b_i^\dagger b_j f_{i\sigma}^\dagger f_{j\sigma} + \text{h.c.} \right). \quad (\text{S4})$$

Introducing the bond fields

$$\chi_{ij}^b \equiv \langle b_i^\dagger b_j \rangle = |\chi_{ij}^b| e^{i\phi_{ij}^b} \quad (\text{S5a})$$

$$\chi_{ij}^f \equiv \sum_{\sigma} \langle f_{i\sigma}^\dagger f_{j\sigma} \rangle = |\chi_{ij}^f| e^{i\phi_{ij}^f}. \quad (\text{S5b})$$

After mean-field decoupling, the kinetic Hamiltonian (S4) becomes

$$H_f^{(0)} = -t \sum_{ij, \sigma} e^{i\bar{A}_{ij}} \chi_{ij}^b f_{i\sigma}^\dagger f_{j\sigma} + \text{h.c.} + h \sum_i n_i^f \quad (\text{S6a})$$

$$H_b^{(0)} = -t \sum_{ij} e^{i\bar{A}_{ij}} \chi_{ij}^f b_i^\dagger b_j + \text{h.c.} + \frac{U}{2} \sum_i L_i^2 + h \sum_i (L_i - 1) \quad (\text{S6b})$$

where the constant term is ignored. For the translation-invariant saddle, the Lagrange multiplier can be relaxed to a uniform number h . The slave-rotor decomposition introduces an internal $U(1)$ gauge redundancy $f_{i\sigma} \rightarrow e^{i\alpha_i} f_{i\sigma}$ and $b_i \rightarrow e^{-i\alpha_i} b_i$. Note that under the internal gauge transformation, the bond phases transform as $\phi_{ij}^b \rightarrow \phi_{ij}^b + \alpha_i - \alpha_j$ and $\phi_{ij}^f \rightarrow \phi_{ij}^f - \alpha_i + \alpha_j$. The individual bond phases are therefore gauge dependent, whereas their sum and the corresponding loop fluxes are gauge invariant. Before imposing a particular saddle, from Eq. (S6) we define the two static phases that actually enter the parton hopping Hamiltonians,

$$\vartheta_{ij}^f \equiv \bar{A}_{ij} + \phi_{ij}^b, \quad \vartheta_{ij}^b \equiv \bar{A}_{ij} + \phi_{ij}^f. \quad (\text{S7})$$

Thus, ϑ_{ij}^f is the effective hopping phase seen by the spinon, whereas ϑ_{ij}^b is the effective hopping phase seen by the char-

gon. Equivalently,

$$t_{ij}^f e^{i\vartheta_{ij}^f} \equiv t e^{i\bar{A}_{ij}} \chi_{ij}^b, \quad t_{ij}^b e^{i\vartheta_{ij}^b} \equiv t e^{i\bar{A}_{ij}} \chi_{ij}^f, \quad (\text{S8})$$

with $t_{ij}^f = t |\chi_{ij}^b|$ and $t_{ij}^b = t |\chi_{ij}^f|$. In particular, $\bar{A}_{ij} + \phi_{ij}^b + \phi_{ij}^f$ controls the factorized physical bond current, and a generic finite-field saddle may support equilibrium loop currents.

The mean-field equations determine the bond fields $\chi_{ij}^{b/f}$ with fixed $\bar{A}_{ij}$; they do not determine the external flux $\bar{A}_{ij}$. Within this fixed commensurate model, we adopt the translation-invariant flux-partition ansatz used for the $SU(2)$ CSL in Refs. [34, 35],

$$\sum_{\partial\Delta} \vartheta_{ij}^f = \frac{\pi}{2}, \quad \sum_{\partial\Delta} \vartheta_{ij}^b = 0 \pmod{2\pi}. \quad (\text{S9})$$

This ansatz is a self-consistent saddle branch rather than a consequence of gauge redundancy or a unique solution for arbitrary external flux. A zero-flux rotor bond correlator leaves the spinon hopping with the microscopic Hofstadter phase; conversely, the filled spinon Hofstadter band produces a bond field whose phase cancels $\bar{A}_{ij}$ in the rotor hopping. The mean-field equations then determine the corresponding hopping magnitudes and the remaining saddle parameters.

Because the chargon link configuration has zero flux, it is pure gauge in the bulk. Any pattern of π link signs compatible with zero flux on every contractible loop is likewise pure gauge and can be removed. By contrast, assigning a phase π to every bond of the nonbipartite triangular lattice produces a π flux through each triangle and belongs to a different flux sector. We choose the convenient bond gauge

$$\vartheta_{ij}^f = \bar{A}_{ij}, \quad \vartheta_{ij}^b = 0, \quad (\text{S10})$$

which, by Eq. (S7), is equivalent to

$$\phi_{ij}^b = 0, \quad \phi_{ij}^f = -\bar{A}_{ij}. \quad (\text{S11})$$

The factor $e^{i\bar{A}_{ij}}$ in the chargon hopping is therefore canceled by the spinon bond field χ_{ij}^f , while the real rotor bond field χ_{ij}^b leaves the spinon hopping with the full microscopic Peierls phase. The relation $\bar{A}_{ij} + \phi_{ij}^b + \phi_{ij}^f = 0$ and the vanishing factorized bond current are properties of this adopted flux-partition branch.

We next identify the static internal link field and note that the static relative field for chargon vanishes,

$$\bar{a}_{ij} \equiv \vartheta_{ij}^f = \bar{A}_{ij}, \quad \vartheta_{ij}^b = \bar{A}_{ij} - \bar{a}_{ij} = 0. \quad (\text{S12})$$

A fixed mean-field ansatz freezes the phases of the gauge-charged bond fields and therefore amounts to a choice of gauge. To retain the low-energy gauge fluctuations associated with the local parton redundancy, we allow these bond phases to fluctuate about the saddle and parameterize them by a dynamical internal link field a_{ij} . The total internal field

transforms as

$$\bar{a}_{ij} + a_{ij} \rightarrow \bar{a}_{ij} + a_{ij} + \alpha_i - \alpha_j,$$

which makes the separate spinon and chargon Hamiltonians gauge covariant. We next restore the dynamical internal fluctuation $a_{ij} = \int_i^j \mathbf{a} \cdot d\mathbf{r}$ and the optical probe $A_{ij} = \int_i^j \mathbf{A} \cdot d\mathbf{r}$, and define $\mathcal{A}_{ij} \equiv A_{ij} - a_{ij}$. Since $\bar{a}_{ij} = \bar{A}_{ij}$ at the adopted saddle, the full phase seen by the chargon reduces to

$$(\bar{A}_{ij} + A_{ij}) - (\bar{a}_{ij} + a_{ij}) = A_{ij} - a_{ij} = \mathcal{A}_{ij}.$$

The two parton Hamiltonians expanded around this saddle are therefore

$$H_f = - \sum_{ij,\sigma} t_{ij}^f e^{i(\bar{A}_{ij} + a_{ij})} f_{i\sigma}^\dagger f_{j\sigma} + \text{h.c.} + h \sum_i n_i^f, \quad (\text{S13})$$

$$H_b = - \sum_{ij} t_{ij}^b e^{i\mathcal{A}_{ij}} b_i^\dagger b_j + \text{h.c.} + \frac{U}{2} \sum_i L_i^2 + h \sum_i (L_i - 1). \quad (\text{S14})$$

where we have utilized condition (S12). Here, h enforces Eq. (S3) on average; its half-filled saddle-point value vanishes in the particle-hole-symmetric rotor problem ($h = 0$). From the above two Hamiltonian we see that the spinon inherits the full microscopic Hofstadter flux from the static external magnetic field. Additionally, the unbar fields a_{ij} and A_{ij} can now be understood as the fluctuating fields with respect to the static background in a unified manner (A_{ij} corresponds to the probe field, which is also a kind of perturbation to the vacuum). The spinon therefore couples to $\bar{a} + a = \bar{A} + a$, whereas the chargon couples only to the fluctuating relative field $\mathcal{A} = A - a$ (its static relative background vanishes). The product $c = bf$ is invariant under the internal gauge transformations and has the physical electromagnetic charge. In the microscopic saddle above, the two parton sectors have different topology because they experience different static fluxes, while their hopping magnitudes t^f and t^b are fixed by different bond expectation values.

The difference in static backgrounds is precisely what allows the two parton sectors to have different topology. The spinons retain $\sum_{\partial\Delta} \bar{a}_{ij} = \pi/2$ and form a spin-singlet Chern insulator with $C_s = 1$ per spin flavor, hence total spinon Chern number $C_f = 2$. The chargon hopping, by contrast, has a real zero-flux background and is topologically trivial at the mean-field level.

We finally clarify the role of the Zeeman coupling. Equation (S1) defines the spin-independent, or orbital-only, Hofstadter-Hubbard model: the static magnetic background enters through its Peierls phase, while the two $SU(2)$ flavors remain exactly degenerate. This is the appropriate effective description when the flavors are Zeeman-insensitive pseudospins (such as layer or valley labels) or when the orbital flux is generated synthetically. If $\sigma = \uparrow, \downarrow$ instead denotes the physical electron spin in a real magnetic field, the symmetry-

allowed microscopic Hamiltonian also contains

$$H_Z = - \frac{\Delta_Z}{2} \sum_i (n_{i\uparrow} - n_{i\downarrow}), \quad \Delta_Z = g_{\text{eff}} \mu_B |\bar{\mathbf{B}}|. \quad (\text{S15})$$

Here the orbital flux is controlled by the perpendicular component of $\bar{\mathbf{B}}$, whereas Δ_Z is controlled by the total field and the effective g factor. These are consequently distinct parameters of the low-energy lattice model.

The Zeeman term does not couple directly to the rotor as in the on-site terms, chargon phase cancels each other and the Zeeman term becomes

$$H_Z = - \frac{\Delta_Z}{2} \sum_i (f_{i\uparrow}^\dagger f_{i\uparrow} - f_{i\downarrow}^\dagger f_{i\downarrow}), \quad (\text{S16})$$

In the magnetic-sublattice basis, the Hamiltonian of each spin flavor is therefore

$$h_{f,\sigma}(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\tau} - \frac{\sigma \Delta_Z}{2} \tau_0.$$

The Zeeman term changes only the scalar τ_0 component: it does not change the vector $\mathbf{d}(\mathbf{k})$, its magnitude $d(\mathbf{k})$, or the current vertex $\partial_{k_i} h_{f,\sigma}$. It merely shifts the two spinon spectra as $E_{\lambda\sigma}(\mathbf{k}) = E_\lambda(\mathbf{k}) - \sigma \Delta_Z/2$. The Zeeman field consequently reduces both the mean-field single-spinon gap to the chemical potential and the energy of a spin-flip excitation. These gaps should not, however, be identified with the interband threshold entering the present optical kernel. The orbital current vertex is spin conserving, $J_i = \partial_{k_i} h_f \otimes \mathbf{1}_\sigma$, and therefore connects only $(-, \sigma)$ to $(+, \sigma)$. For every such transition,

$$\begin{aligned} E_{+\sigma}(\mathbf{k}) - E_{-\sigma}(\mathbf{k}) \\ = \left[E_+(\mathbf{k}) - \frac{\sigma \Delta_Z}{2} \right] - \left[E_-(\mathbf{k}) - \frac{\sigma \Delta_Z}{2} \right] = 2d(\mathbf{k}). \end{aligned}$$

Thus the Zeeman shift cancels between the initial and final states. As long as Δ_Z is smaller than the zero-field spin-flip (or indirect spinon) gap, one lower $C_s = 1$ band remains completely occupied for each spin flavor. There is then no transfer of occupation between the two flavors, the spin-summed bond field χ_{ij}^f is unchanged, and the chargon Hamiltonian remains the real, zero-flux Hamiltonian in Eq. (S14). At zero temperature, the Fermi occupation factors, current matrix elements, and spin-conserving transition energies entering $\Pi_f^{ij}(\omega)$ are consequently identical to their $\Delta_Z = 0$ values within this subcritical regime. In the interacting language the corresponding condition is $\Delta_Z < \Delta_{\text{spin}}$, with Δ_{spin} the zero-field gap to the lowest spinful excitation. The state then remains adiabatically connected to the Kalmeyer-Laughlin CSL, although a nonzero Δ_Z explicitly reduces spin $SU(2)$ to $U(1)$.

Once the Zeeman splitting closes the spin gap, spinon Fermi pockets, magnetization, and a reconstruction of the internal-flux saddle can occur; such field-induced phases have been analyzed in Ref. [33]. They lie outside the gapped, spin-degenerate $C_f = 2$ regime considered here. Throughout this

work we therefore set $\Delta_Z = 0$, while the response theory also applies without modification for $\Delta_Z < \Delta_{\text{spin}}$ under the conditions just stated. In particular, the Zeeman term neither acts on the chargon directly nor changes the chargon antiunitary constraint discussed later.

TENSOR IOFFE–LARKIN RULE

We now derive the physical electromagnetic response around the saddle constructed in the preceding section. The static backgrounds $\bar{A}$ and $\bar{a} = \bar{A}$ are already included in the mean-field band Hamiltonians. Throughout this section, the unbarred fields A_μ and a_μ denote fluctuations about those backgrounds, and

$$\mathcal{A}_\mu \equiv A_\mu - a_\mu \quad (\text{S17})$$

is the fluctuating relative field minimally coupled to the chargon.

In the Mott phase the rotor is uncondensed, $\langle b_i \rangle = \langle e^{i\theta_i} \rangle = 0$, and the chargon sector is gapped. Its response to $\mathcal{A}_\mu$ is defined by

$$e^{-S_b^{\text{eff}}[\mathcal{A}]} = \int \mathcal{D}b \mathcal{D}b^* e^{-S_b[b; \mathcal{A}]}. \quad (\text{S18})$$

Expanding about the stationary zero-probe saddle, the first nontrivial term is quadratic,

$$S_b^{\text{eff}}[\mathcal{A}] = \frac{1}{2} \sum_q \mathcal{A}_\mu(-q) \Pi_b^{\mu\nu}(q) \mathcal{A}_\nu(q) + \dots, \quad (\text{S19})$$

where $q = (i\Omega_n, \mathbf{q})$ and

$$\Pi_b^{\mu\nu}(q) = \left. \frac{\delta^2 S_b^{\text{eff}}[\mathcal{A}]}{\delta \mathcal{A}_\mu(-q) \delta \mathcal{A}_\nu(q)} \right|_{\mathcal{A}=0} \quad (\text{S20})$$

is the irreducible chargon polarization kernel. Because the chargon is gapped, this kernel is analytic at low frequency and long wavelength and describes virtual charge polarization, including virtual doublon–holon fluctuations, of the Mott insulator.

The gapped spinons form a Chern insulator and couple minimally to a_μ . Integrating them out gives

$$S_f^{\text{eff}}[a] = \frac{1}{2} \sum_q a_\mu(-q) \Pi_f^{\mu\nu}(q) a_\nu(q) + \dots \quad (\text{S21})$$

The kernel $\Pi_f^{\mu\nu}$ contains both the paramagnetic current–current correlator and the diamagnetic contribution. In the CSL it also contains the Chern–Simons response and the finite-frequency interband response of the spinon Chern bands.

For completeness, the temporal component of the internal gauge field is the fluctuation of the Lagrange multiplier that

enforces Eq. (S3). Writing

$$\lambda_i(\tau) = h + a_{0,i}(\tau) \quad (\text{S22})$$

completes the dynamical field $a_\mu = (a_0, \mathbf{a})$: the spatial components are the bond-phase fluctuations discussed in the preceding section, while a_0 enforces the local constraint. The half-filled particle–hole-symmetric rotor saddle has $h = 0$. In the uniform finite-frequency optical limit, we choose temporal gauge,

$$A_0 = a_0 = 0, \quad (\text{S23})$$

so that the density sector decouples and only $i, j \in \{x, y\}$ remain. The total effective action up to quadratic order then reads

$$S^{\text{eff}}[A, a] = S_b^{\text{eff}} + S_f^{\text{eff}} = \frac{1}{2} \sum_{\Omega_n} \mathcal{A}_i \Pi_b^{ij} \mathcal{A}_j + \frac{1}{2} \sum_{\Omega_n} a_i \Pi_f^{ij} a_j, \quad (\text{S24})$$

where the frequency argument Ω_n is suppressed for notational simplicity. Since a_i is an internal gauge fluctuation, it should also be integrated out from the physical response. At the Gaussian level this is equivalent to solving its saddle-point equation $\delta S^{\text{eff}}/\delta a_i = 0$, which gives

$$\mathbf{a} = [\Pi_b + \Pi_f]^{-1} \Pi_b \mathbf{A} \quad (\text{S25})$$

This matrix equation gives the internal gauge field induced by the physical vector potential; spatial indices and frequency arguments are suppressed. Substituting the saddle-point solution back into Eq. (S24) gives the effective action for the physical electromagnetic field, which we denote as $S_{\text{phys}}[A]$:

$$S_{\text{phys}}[A] = \frac{1}{2} \sum_{\Omega_n} A_i(-\Omega_n) \Pi_{\text{phys}}^{ij}(i\Omega_n) A_j(\Omega_n). \quad (\text{S26})$$

Here the physical electromagnetic response kernel is

$$\Pi_{\text{phys}} = \Pi_b - \Pi_b [\Pi_b + \Pi_f]^{-1} \Pi_b. \quad (\text{S27})$$

All three kernels are matrices in spatial indices. To make contact with the standard tensor Ioffe–Larkin rule, we first rewrite Eq. (S27) as

$$\begin{aligned} \Pi_{\text{phys}} &= \Pi_b [I - (\Pi_b + \Pi_f)^{-1} \Pi_b] \\ &= \Pi_b [(\Pi_b + \Pi_f)^{-1} (\Pi_b + \Pi_f) - (\Pi_b + \Pi_f)^{-1} \Pi_b] \\ &= \Pi_b (\Pi_b + \Pi_f)^{-1} \Pi_f. \end{aligned} \quad (\text{S28})$$

When the relevant kernels are invertible, taking the matrix inverse gives

$$\begin{aligned} \Pi_{\text{phys}}^{-1} &= \Pi_f^{-1} (\Pi_b + \Pi_f) \Pi_b^{-1} \\ &= \Pi_f^{-1} + \Pi_b^{-1}. \end{aligned} \quad (\text{S29})$$

Thus, the physical response obeys the well-known tensor

Ioffe–Larkin composition rule

$$\Pi_{\text{phys}}^{-1} = \Pi_b^{-1} + \Pi_f^{-1} \quad (\text{S30})$$

without requiring Π_b and Π_f to commute. At frequencies where an individual kernel is singular, Eq. (S27) remains the more direct form, while Eq. (S30) is understood by a regulated limit. The measured response is therefore neither the bare spinon kernel nor the bare chargon kernel, but their gauge-constrained composition.

We now show why the chargon kernel has no antisymmetric, or Hall, component at the adopted saddle. Because the static relative flux vanishes, the hopping amplitudes in H_b are real. At half filling $h = 0$, and complex conjugation in the rotor basis defines an antiunitary operation Θ_b satisfying

$$\begin{aligned} \Theta_b b_i \Theta_b^{-1} &= b_i^\dagger, \\ \Theta_b L_i \Theta_b^{-1} &= -L_i, \\ \Theta_b H_b[\mathcal{A}] \Theta_b^{-1} &= H_b[-\mathcal{A}]. \end{aligned} \quad (\text{S31})$$

Thus $H_b[0]$ and its equilibrium density matrix are invariant under Θ_b . The uniform chargon current $j_b^i = -\partial H_b / \partial \mathcal{A}_i|_{\mathcal{A}=0}$ is odd,

$$\Theta_b j_b^i \Theta_b^{-1} = -j_b^i, \quad (\text{S32})$$

whereas the diamagnetic contact tensor, obtained from the second derivative of H_b with respect to $\mathcal{A}$, is even and symmetric in its spatial indices.

To see the resulting reciprocity explicitly, consider the spectral representation of the retarded paramagnetic kernel,

$$\Pi_{b,\text{para}}^{R,ij}(\omega) = \sum_{m,n} \frac{p_m - p_n}{\omega + E_m - E_n + i0^+} \langle m | j_b^i | n \rangle \langle n | j_b^j | m \rangle, \quad (\text{S33})$$

where $p_m = e^{-\beta E_m} / Z$. Antiunitarity and the odd transformation of the current imply

$$\langle \Theta_b m | j_b^i | \Theta_b n \rangle = -\langle m | j_b^i | n \rangle^*. \quad (\text{S34})$$

The states $|m\rangle$ and $\Theta_b |m\rangle$ have the same energy and Boltzmann weight. Relabeling every state in Eq. (S33) by its antiunitary partner therefore exchanges the two current matrix elements. Since both currents are odd, their two minus signs cancel, giving the Onsager relation

$$\Pi_b^{R,ij}(\omega) = \Pi_b^{R,ji}(\omega). \quad (\text{S35})$$

The symmetric diamagnetic contact term obeys the same relation, so it applies to the full kernel and it holds for the Matsubara kernel before analytic continuation. Hence, in the uniform optical limit,

$$\Pi_b^H(\omega) \equiv \frac{\Pi_b^{R,xy}(\omega) - \Pi_b^{R,yx}(\omega)}{2} = 0. \quad (\text{S36})$$

This antiunitary symmetry belongs to the real, zero-relative-

flux chargon Hamiltonian. It is not a time-reversal symmetry of the microscopic Hofstadter–Hubbard model, whose external flux $\Phi_\Delta = \pi/2$ explicitly breaks time reversal. Importantly, the argument forbids the entire finite-frequency chargon Hall kernel, not only its dc chargon Hall kernel.

GAUSSIAN CHARGON RESPONSE FROM THE SLAVE-ROTOR ACTION

We now derive the chargon kernel directly from the Gaussian slave-rotor theory with half filling, where the uniform saddle has $h = 0$ and temporal gauge $\mathcal{A}_0 = 0$ is used. Starting from chargon Hamiltonian [Eq. (S14)], the canonical Euclidean rotor action is

$$\begin{aligned} S_b[L, \theta; \mathcal{A}] &= \int_0^\beta d\tau \left[\sum_i \left(iL_i \partial_\tau \theta_i + \frac{U}{2} L_i^2 \right) \right. \\ &\quad \left. - \sum_{\langle ij \rangle} t_{ij}^b \left(e^{i\mathcal{A}_{ij}} e^{-i\theta_i} e^{i\theta_j} + \text{c.c.} \right) \right]. \end{aligned} \quad (\text{S37})$$

The first term is the canonical, or Berry-phase-like term of the rotor. Since θ_i and L_i are conjugate variables, $[\theta_i, L_j] = i\delta_{ij}$, so the phase-space path integral for $Z = \text{Tr} e^{-\beta H_b}$ has the Euclidean form $S_E = \int d\tau [iL_i \partial_\tau \theta_i + H_b]$. Within the Gaussian rotor approximation the integer-valued L_i is treated as a continuous field. Completing the square

$$iL_i \partial_\tau \theta_i + \frac{U}{2} L_i^2 = \frac{U}{2} \left(L_i + \frac{i}{U} \partial_\tau \theta_i \right)^2 + \frac{1}{2U} (\partial_\tau \theta_i)^2.$$

and integrating it out gives the action [31]

$$\begin{aligned} S_b[\theta; \mathcal{A}] &= \int_0^\beta d\tau \left[\frac{1}{2U} \sum_i (\partial_\tau \theta_i)^2 \right. \\ &\quad \left. - \sum_{\langle ij \rangle} t_{ij}^b \left(e^{i\mathcal{A}_{ij}} e^{-i\theta_i} e^{i\theta_j} + \text{c.c.} \right) \right]. \end{aligned} \quad (\text{S38})$$

Because θ_i is an angular variable, θ_i and $\theta_i + 2\pi$ describe the same rotor. An exact finite-temperature path integral must therefore also sum over integer winding sectors satisfying $\theta_i(\beta) = \theta_i(0) + 2\pi m_i$. Here we retain the smooth Gaussian fluctuations and ignore the nonperturbative winding events. To obtain a quadratic theory, define $X_i = e^{i\theta_i}$. For the exact unit-modulus field we have

$$|X_i|^2 = 1, \quad |\partial_\tau X_i|^2 = (\partial_\tau \theta_i)^2, \quad (\text{S39})$$

The soft-rotor approximation then relaxes the pointwise condition $|X_i|^2 = 1$ to an average condition and imposes it with a

uniform multiplier λ . The resulting quadratic action is

$$S_b^G[X; \mathcal{A}] = \int_0^\beta d\tau \left[\frac{1}{2U} \sum_i |\partial_\tau X_i|^2 + \lambda \sum_i (|X_i|^2 - 1) - \sum_{\langle ij \rangle} t_{ij}^b (e^{i\mathcal{A}_{ij}} X_i^* X_j + \text{c.c.}) \right]. \quad (\text{S40})$$

For a translation-invariant, real zero-flux hopping pattern, define

$$\varepsilon_b(\mathbf{k}) = 2 \sum_\delta t_\delta^b \cos(\mathbf{k} \cdot \delta), \quad \xi_b(\mathbf{k}) = \lambda - \varepsilon_b(\mathbf{k}) > 0, \quad (\text{S41})$$

where one vector from each undirected bond pair is included in the δ sum. The $\xi_b(\mathbf{k})$ term is positive definite, and the rotor is uncondensed, when $\lambda > \max_{\mathbf{k}} \varepsilon_b(\mathbf{k})$ or equivalently $\xi_b(\mathbf{k}) > 0$. At equality the lowest rotor mode becomes gapless and can condense. Thus the Mott saddle has the single-chargon gap $\Delta_b = \min_{\mathbf{k}} E_b(\mathbf{k}) > 0$. At zero probe field, use the bosonic Fourier transform

$$X_i(\tau) = \frac{1}{\sqrt{\beta N}} \sum_{\mathbf{k}, v_n} e^{i\mathbf{k} \cdot \mathbf{r}_i - i v_n \tau} X(\mathbf{k}, i v_n), \quad v_n = 2\pi n T. \quad (\text{S42})$$

Substitution into Eq. (S40) diagonalizes the quadratic action:

$$S_b^G[\mathcal{A} = 0] = \sum_{\mathbf{k}, v_n} X^*(\mathbf{k}, i v_n) \left[\frac{v_n^2}{2U} + \xi_b(\mathbf{k}) \right] X(\mathbf{k}, i v_n) - \beta N \lambda. \quad (\text{S43})$$

For a Gaussian complex field, its two-point Green's function is the inverse of the quadratic coefficient. Before evaluating it, we implement the atomic-limit calibration of the soft-rotor approximation. For the exact compact rotor, $H_{\text{at}} = UL^2/2$, the $L = 0 \rightarrow L = \pm 1$ excitation costs $U/2$. By contrast, relaxing $|X_i| = 1$ to the average constraint in Eq. (S46) would place the atomic pole at U if the same U were used directly in the Gaussian propagator. We therefore replace the inertial parameter of the soft theory by $U_s \equiv U/2$, while U continues to denote the physical Hubbard interaction in the exact rotor Hamiltonian and phase-only action above. Equivalently, the U in the soft kinetic term of Eq. (S43) is understood as U_s when constructing the Gaussian propagator and all response functions below. Thus,

$$G_b(\mathbf{k}, i v_n) \equiv \langle X(\mathbf{k}, i v_n) X^*(\mathbf{k}, i v_n) \rangle = \frac{1}{v_n^2 / (2U_s) + \xi_b(\mathbf{k})} = \frac{2U_s}{v_n^2 + E_b^2(\mathbf{k})}, \quad (\text{S44})$$

$$E_b(\mathbf{k}) = \sqrt{2U_s \xi_b(\mathbf{k})}. \quad (\text{S45})$$

Varying the Gaussian free energy with respect to λ restores the relaxed unit-modulus condition on average:

$$1 = \langle |X_i|^2 \rangle = \frac{T}{N} \sum_{\mathbf{k}, v_n} G_b(\mathbf{k}, i v_n). \quad (\text{S46})$$

This constraint also makes the atomic-limit calibration ex-

plicit. When $t_b = 0$ ($\varepsilon_b = 0$), the rotor mode is momentum independent, $E_b(\mathbf{k}) = E_0 = \sqrt{2U_s \lambda}$. At zero temperature, the elementary bosonic frequency sum gives

$$1 = T \sum_{v_n} G_b(i v_n) = \frac{U_s}{E_0} \rightarrow E_0 = U_s, \quad \lambda = \frac{U_s}{2} \quad (\text{S47})$$

Hence the atomic single-chargon gap is $\Delta_b = E_0 = U_s = U/2$, not $U_s/2$. The latter is the atomic value of the soft constraint multiplier λ , rather than an excitation energy. At large but finite U/t_b , the constraint has no correction linear in t_b to λ , because the Brillouin-zone average of $\varepsilon_b(\mathbf{k})$ vanishes. Using $\Delta_b = \sqrt{2U_s(\lambda - 6t_b)}$ therefore gives

$$\Delta_b = U_s - 6t_b + O(t_b^2/U_s) = \frac{U}{2} - 6t_b + O(t_b^2/U). \quad (\text{S48})$$

Thus $U/2$ is the leading atomic-limit value, while the experimentally calibrated gap should include the finite-bandwidth renormalization.

In a fully self-consistent treatment, Eq. (S46), together with the bond saddle equations, determines λ and t_{ij}^b . In the present work, we instead take an uncondensed CSL saddle as input and focus on probing the spinon QGT in the main text.

We next calculate the uniform optical response. A spatially uniform, time-dependent relative vector potential enters by the Peierls substitution $\xi_b(\mathbf{k}) \rightarrow \xi_b(\mathbf{k} - \mathcal{A})$. Introduce the vertices

$$v_i(\mathbf{k}) = \partial_{k_i} \xi_b(\mathbf{k}), \quad m_{ij}(\mathbf{k}) = \partial_{k_i} \partial_{k_j} \xi_b(\mathbf{k}). \quad (\text{S49})$$

For completeness, the expansion leading to the response kernel is

$$\xi_b(\mathbf{k} - \mathcal{A}) = \xi_b(\mathbf{k}) - v_i(\mathbf{k}) \mathcal{A}_i + \frac{1}{2} m_{ij}(\mathbf{k}) \mathcal{A}_i \mathcal{A}_j + O(\mathcal{A}^3). \quad (\text{S50})$$

Writing $G_b^{-1}[\mathcal{A}] = G_b^{-1} + V_1 + V_2 + \dots$, where V_1 and V_2 are respectively the terms linear and quadratic in $\mathcal{A}$, integrating out the complex field gives

$$S_{\text{eff}}[\mathcal{A}] = \text{Tr} \ln G_b^{-1} + \text{Tr}(G_b V_1) + \text{Tr}(G_b V_2) - \frac{1}{2} \text{Tr}(G_b V_1 G_b V_1) + O(\mathcal{A}^3). \quad (\text{S51})$$

The first-order term vanishes after the Brillouin-zone sum. The $\text{Tr}(G_b V_2)$ term produces the diamagnetic contact vertex m_{ij} , whereas the term with two V_1 's produces the paramagnetic bubble with vertices $v_i v_j$. Defining $S_{\text{eff}}^{(2)} = \frac{1}{2} \sum_{\Omega_n} \mathcal{A}_i (-i\Omega_n) \Pi_{b,E}^{ij}(i\Omega_n) \mathcal{A}_j (i\Omega_n)$ then gives

$$\Pi_{b,E}^{ij}(i\Omega_n) = \int_{\mathbf{k}} T \sum_{v_n} [m_{ij}(\mathbf{k}) G_b(\mathbf{k}, i v_n) - v_i(\mathbf{k}) v_j(\mathbf{k}) G_b(\mathbf{k}, i v_n) G_b(\mathbf{k}, i v_n + i\Omega_n)], \quad (\text{S52})$$

where $\int_{\mathbf{k}} \equiv N^{-1} \sum_{\mathbf{k}}$. This expression is already the complete Gaussian chargon kernel; omitting the first term would violate

gauge invariance. Indeed, periodicity of the Brillouin zone gives

$$\Pi_{b,E}^{ij}(0) = \int_{\mathbf{k}} T \sum_{v_n} \partial_{k_i} [v_j(\mathbf{k}) G_b(\mathbf{k}, i\nu_n)] = 0. \quad (\text{S53})$$

Thus a static uniform vector potential produces no current in the Mott insulator.

At zero temperature the Matsubara frequency sums are

$$T \sum_{v_n} G_b = \frac{U_s}{E_b},$$

$$T \sum_{v_n} G_b(i\nu_n) G_b(i\nu_n + i\Omega_n) = \frac{4U_s^2}{E_b(\Omega_n^2 + 4E_b^2)}. \quad (\text{S54})$$

An integration by parts converts the contact term according to

$$\int_{\mathbf{k}} \frac{U_s}{E_b} m_{ij} = \int_{\mathbf{k}} \frac{U_s^2}{E_b^3} v_i v_j. \quad (\text{S55})$$

Consequently, the Euclidean kernel takes the manifestly gauge-invariant form

$$\Pi_{b,E}^{ij}(i\Omega_n) = \int_{\mathbf{k}} \frac{U_s^2 v_i(\mathbf{k}) v_j(\mathbf{k})}{E_b^3(\mathbf{k})} \frac{\Omega_n^2}{\Omega_n^2 + 4E_b^2(\mathbf{k})}. \quad (\text{S56})$$

Writing $z = \omega + i0^+$, the same analytic continuation used for the spinon kernel gives

$$\Pi_b^{R,ij}(\omega) = \int_{\mathbf{k}} \frac{U_s^2 v_i(\mathbf{k}) v_j(\mathbf{k})}{E_b^3(\mathbf{k})} \frac{z^2}{z^2 - 4E_b^2(\mathbf{k})}. \quad (\text{S57})$$

The optical threshold is therefore $2\Delta_b$: below it the clean Gaussian kernel is real and describes only virtual chargon-antichargon polarization, whereas above it the imaginary part resolves the continuum of two-chargon excitations.

We now apply the result to the triangular lattice with three undirected nearest-neighbor vectors $\delta_1 = a(1, 0)$, $\delta_2 = a(1/2, \sqrt{3}/2)$, $\delta_3 = a(-1/2, \sqrt{3}/2)$. Although the physical system is placed in an external orbital field, the Hofstadter–Hubbard saddle chosen above obeys $\bar{A}_{ij} = \bar{A}_{ji} - \bar{a}_{ij} = 0$. Its hopping is therefore real and translation invariant in the original one-site unit cell. The doubled magnetic unit cell associated with the spinon Hofstadter problem does not enter the chargon dispersion. For a uniform nearest-neighbor amplitude $t_b > 0$, Eqs. (S41) and (S45) become

$$\gamma_{\Delta}(\mathbf{k}) = \cos(ak_x) + 2 \cos\left(\frac{ak_x}{2}\right) \cos\left(\frac{\sqrt{3}ak_y}{2}\right),$$

$$\varepsilon_b(\mathbf{k}) = 2t_b \gamma_{\Delta}(\mathbf{k}),$$

$$\xi_b(\mathbf{k}) = r + 2t_b[3 - \gamma_{\Delta}(\mathbf{k})],$$

$$E_b(\mathbf{k}) = \sqrt{2U_s \xi_b(\mathbf{k})},$$

where $r \equiv \lambda - 6t_b > 0$ and $\Delta_b = \sqrt{2U_s r}$. Note that ε_b is maximal at Γ , while $\varepsilon_{b,\min} = -3t_b$ at the K points. Thus

$$E_{b,\max} = \sqrt{\Delta_b^2 + 18U_s t_b}.$$

Equation (S57) generally contains a continuum of pair poles. If the chargon band is narrow compared with its gap, $E_b(\mathbf{k}) \simeq \Delta_b$ over the optically weighted momenta, it reduces to an effective single pole. To see this directly, first rewrite each term in Eq. (S57) exactly as

$$\Pi_b^{R,ij}(z) = -z^2 \int_{\mathbf{k}} w_{ij}(\mathbf{k}) \frac{4E_b^2(\mathbf{k})}{4E_b^2(\mathbf{k}) - z^2},$$

$$w_{ij}(\mathbf{k}) \equiv \frac{U_s^2 v_i(\mathbf{k}) v_j(\mathbf{k})}{4E_b^5(\mathbf{k})}, \quad \chi_b^{ij} = \int_{\mathbf{k}} w_{ij}(\mathbf{k}). \quad (\text{S58})$$

Writing $E_b(\mathbf{k}) = \Delta_b + \delta E_b(\mathbf{k})$ and denoting the chargon bandwidth by W_b , the narrow-band condition is $W_b/\Delta_b \ll 1$ or $\delta E_b(\mathbf{k})/\Delta_b \ll 1$. Thus, we have the approximation

$$\frac{4E_b^2(\mathbf{k})}{4E_b^2(\mathbf{k}) - z^2} \approx \frac{4\Delta_b^2}{4\Delta_b^2 - z^2}. \quad (\text{S59})$$

Note that the chargon band gap is larger than spinon band gap and when on-shell resonance of spinon kernel occurs, the spinon frequency is below the chargon threshold, away from the chargon pole. Therefore, under the narrow-band condition, we have

$$\Pi_b^{R,ij}(\omega) \simeq -\chi_b^{ij} z^2 \frac{4\Delta_b^2}{4\Delta_b^2 - z^2}. \quad (\text{S60})$$

The approximation is controlled throughout a subgap spinon spectroscopy window provided both $W_b/\Delta_b \ll 1$ and $|4\Delta_b^2 - z^2|$ remains large compared with the bandwidth correction. It necessarily fails parametrically close to the chargon-pair threshold, where the individual continuum poles can no longer be replaced by one pole. Eq. (S60) has the same form of the response kernel of a simple anisotropically polarizable medium after identifying its optical pole $\Omega_b = 2\Delta_b$. The minus sign in Eq. (S60) is the retarded vector-potential-kernel convention obtained by analytic continuation of the Euclidean action; it must be used consistently when comparing with the phenomenological expression.

Importantly, the leading low-frequency response does not require a narrow chargon band. Expanding the full Gaussian continuum in Eq. (S57) gives

$$\Pi_b^{R,ij}(z) = -\chi_b^{ij} z^2 - \gamma_b^{ij} z^4 + O(z^6) \quad (\text{S61})$$

$$\gamma_b^{ij} = \int_{\mathbf{k}} \frac{U_s^2 v_i(\mathbf{k}) v_j(\mathbf{k})}{16E_b^7(\mathbf{k})}. \quad (\text{S62})$$

By comparison, Eq. (S60) yields $\Pi_b^{R,ij} = -\chi_b^{ij} z^2 - \chi_b^{ij} z^4/(4\Delta_b^2) + O(z^6)$. The narrow-band condition $E_b(\mathbf{k}) \simeq \Delta_b$ makes $\gamma_b^{ij} \simeq \chi_b^{ij}/(4\Delta_b^2)$ and hence controls the single-pole form beyond leading order. Without that condition, the continuum still has the same universal $-\chi_b^{ij} z^2$ scaling at low frequency. Thus, the low-frequency polarizability χ_b^{ij} is a robust property and can still be extracted from the low-frequency lon-

gitudinal response.

We finally separate the Hall question from spatial isotropy. Apparently, each term in Eqs. (S52) and (S57) is symmetric under $i \leftrightarrow j$, so

$$\Pi_b^{R,ij}(\omega) = \Pi_b^{R,ji}(\omega), \quad \Pi_b^H(\omega) \equiv \frac{\Pi_b^{R,xy} - \Pi_b^{R,yx}}{2} = 0. \quad (\text{S63})$$

This explicitly reproduces the antiunitary-symmetry result of the preceding section. On a completely general anisotropic lattice, however, a symmetric off-diagonal component $\Pi_b^{R,xy} = \Pi_b^{R,yx}$ can be present in a nonprincipal coordinate basis (but it is not a Hall response). The tensor Ioffe–Larkin rule itself requires neither diagonality nor isotropy. The scalar reduction used in the main text follows from the unbroken triangular-lattice rotation symmetry. In the uniform optical limit, let

$$R_3 = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}$$

represent a rotation by $2\pi/3$. Invariance under C_3 imposes the tensor constraint

$$\Pi_b^R(z) = R_3 \Pi_b^R(z) R_3^T.$$

Equation (S63) makes this tensor symmetric, so we may write it as $\Pi_b^R = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$. Substitution into the constraint above gives $a = b$ and $c = 0$. Thus C_3 makes the two longitudinal responses equal, while the absence of a Hall component removes the only other rotationally invariant tensor structure. We therefore define the scalar chargin channel and its non-negative optical weight by

$$\Pi_b^{R,ij}(z) = \Pi_b(z) \delta^{ij}, \quad \chi_b^{ij} = \chi_b \delta^{ij}, \quad (\text{S64a})$$

$$\Pi_b(z) \equiv \frac{1}{2} \delta_{ij} \Pi_b^{R,ij}(z) = -z^2 \int_{\mathbf{k}} w(\mathbf{k}) \frac{4E_b^2(\mathbf{k})}{4E_b^2(\mathbf{k}) - z^2}, \quad (\text{S64b})$$

$$w(\mathbf{k}) \equiv \frac{1}{2} \delta^{ij} w_{ij}(\mathbf{k}) = \frac{U_s^2 [v_x^2(\mathbf{k}) + v_y^2(\mathbf{k})]}{8E_b^5(\mathbf{k})} \geq 0, \quad (\text{S64c})$$

$$\chi_b \equiv \int_{\mathbf{k}} w(\mathbf{k}). \quad (\text{S64d})$$

For any scalar function $O(\mathbf{k})$, the corresponding normalized optical average is

$$\langle O \rangle_w \equiv \frac{\int_{\mathbf{k}} w(\mathbf{k}) O(\mathbf{k})}{\int_{\mathbf{k}} w(\mathbf{k})} = \frac{1}{\chi_b} \int_{\mathbf{k}} w(\mathbf{k}) O(\mathbf{k}). \quad (\text{S65})$$

C_3 requires isotropy only after the Brillouin-zone integral. Specifically, although $w_{ij}(\mathbf{k}) \neq w_{ji}(\mathbf{k})$, after integration over the Brillouin zone, we still have $\chi_b^{xx} = \chi_b^{yy}$ and $\chi_b^{xy} = 0$.

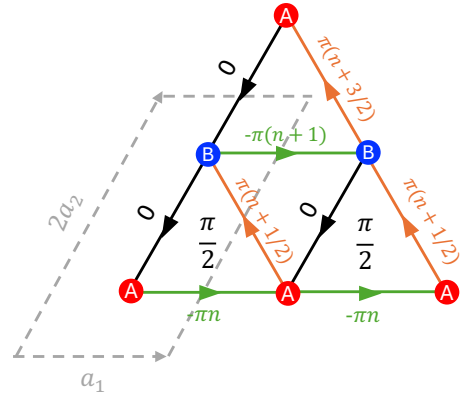


FIG. S1. The magnetic gauge for the $\pi/2$ -flux state on the triangular lattice. The grey dashed lines indicate the magnetic unit cell. Hopping along different bonds with different colors acquires different phases as indicated by the number near the bonds. The magnetic flux threading each triangular plaquette gives a phase $\pi/2$ when traversing the plaquette in a counterclockwise direction.

SPINON HAMILTONIAN OF THE HOFSTADTER–HUBBARD MODEL

In this section we derive the spinon Hamiltonian used in the kernel calculation from the triangular-lattice Hofstadter saddle of Eq. (S13). We work in the uniform optical limit, set the temporal gauge fluctuation to zero, and take a uniform nearest-neighbor spinon hopping t_f . Write the lattice sites as $\mathbf{r}_{m,n} = m\mathbf{a}_1 + n\mathbf{a}_2$, with

$$\mathbf{a}_1 = a(1, 0), \quad \mathbf{a}_2 = a(1/2, \sqrt{3}/2). \quad (\text{S66})$$

For the $\pi/2$ -flux state, a convenient Landau-type magnetic gauge, gauge-equivalent to other commonly used choices for the triangular-lattice Hofstadter problem [34, 35], is

$$\begin{aligned} \bar{a}_{\mathbf{r}, \mathbf{r}+\mathbf{a}_1} &= -\pi n, \\ \bar{a}_{\mathbf{r}, \mathbf{r}+\mathbf{a}_2} &= 0, \\ \bar{a}_{\mathbf{r}, \mathbf{r}+\mathbf{a}_2-\mathbf{a}_1} &= +\pi \left(n + \frac{1}{2} \right) \pmod{2\pi}. \end{aligned} \quad (\text{S67})$$

This choice depends only on the row coordinate n , which makes its two-row magnetic periodicity explicit. It is useful first to verify the flux. Orienting the boundary of each triangle counterclockwise, the upper triangle $\mathbf{r} \rightarrow \mathbf{r} + \mathbf{a}_1 \rightarrow \mathbf{r} + \mathbf{a}_2 \rightarrow \mathbf{r}$ gives

$$\begin{aligned} \Phi_{\triangle \uparrow}^{\text{ccw}} &= \bar{a}_{\mathbf{r}, \mathbf{r}+\mathbf{a}_1} + \bar{a}_{\mathbf{r}+\mathbf{a}_1, \mathbf{r}+\mathbf{a}_2} + \bar{a}_{\mathbf{r}+\mathbf{a}_2, \mathbf{r}} \\ &= -\pi n + \pi \left(n + \frac{1}{2} \right) - 0 = +\frac{\pi}{2}, \end{aligned} \quad (\text{S68})$$

where the second edge is along $\mathbf{a}_2 - \mathbf{a}_1$. For the lower triangle $\mathbf{r} \rightarrow \mathbf{r} + \mathbf{a}_2 \rightarrow \mathbf{r} + \mathbf{a}_2 - \mathbf{a}_1 \rightarrow \mathbf{r}$, one similarly obtains

$$\Phi_{\Delta\downarrow}^{\text{ccw}} = 0 + \pi(n+1) - \pi\left(n + \frac{1}{2}\right) = +\frac{\pi}{2}. \quad (\text{S69})$$

Thus every elementary triangle carries the same uniform flux, with the counterclockwise convention giving $+\pi/2$, consistently with $\partial\Delta$ above. Complex conjugating all bond amplitudes reverses the physical chirality, leaves the dispersion unchanged, and reverses the band Chern number. The gauge in Eq. (S67) is periodic under $n \rightarrow n+2$, so the magnetic primitive vectors may be chosen as $\mathbf{a}_1^m = \mathbf{a}_1$ and $\mathbf{a}_2^m = 2\mathbf{a}_2$. The magnetic unit cell therefore contains two sites, denoted A (n even) and B (n odd), in agreement with the general $N = 2$ Hofstadter counting discussed above.

To see explicitly how the Bloch matrix follows from this gauge, retain one orientation of the three nearest-neighbor bonds, $\delta_1 = \mathbf{a}_1$, $\delta_2 = \mathbf{a}_2$, and $\delta_3 = \mathbf{a}_2 - \mathbf{a}_1$, and write

$$H_f^{(0)} = -t_f \sum_{\mathbf{r}, \sigma} \sum_{\nu=1}^3 (e^{i\bar{\mathbf{a}}_{\nu} \cdot \mathbf{r} + \delta_{\nu}} f_{\mathbf{r}\sigma}^\dagger f_{\mathbf{r}+\delta_{\nu}, \sigma} + \text{h.c.}). \quad (\text{S70})$$

Let $\mathbf{R} = m\mathbf{a}_1 + 2\ell\mathbf{a}_2$ label a magnetic unit cell, with $\mathbf{s}_A = 0$ and $\mathbf{s}_B = \mathbf{a}_2$, and use the full-position Fourier convention

$$f_{\mathbf{R}+\mathbf{s}_\alpha, \sigma} = \frac{1}{\sqrt{N_m}} \sum_{\mathbf{k} \in \text{MBZ}} e^{i\mathbf{k} \cdot (\mathbf{R} + \mathbf{s}_\alpha)} f_{\mathbf{k}\alpha\sigma}. \quad (\text{S71})$$

Introduce $k_1 = \mathbf{k} \cdot \mathbf{a}_1$, $k_2 = \mathbf{k} \cdot \mathbf{a}_2$, and the magnetic-sublattice spinor $\Psi_{\mathbf{k}\sigma} = (f_{\mathbf{k}A\sigma}, f_{\mathbf{k}B\sigma})^T$. For brevity, define $\gamma_1 = \cos k_1$, $\gamma_2 = \cos k_2$, and $\gamma_3 = \cos(k_2 - k_1)$. The $\mathbf{a}_1$ bonds have hopping phases $+1$ on the A rows and -1 on the B rows; the $\mathbf{a}_2$ bonds have phase $+1$; and the $\mathbf{a}_2 - \mathbf{a}_1$ bonds have phases $+i$ and $-i$ when they start, respectively, on A and B . Adding each bond to its Hermitian conjugate therefore gives

$$h_f^{(\mathbf{a}_1)}(\mathbf{k}) = -2t_f \gamma_1 \tau_z, \quad (\text{S72})$$

$$h_f^{(\mathbf{a}_2)}(\mathbf{k}) = -2t_f \gamma_2 \tau_x, \quad (\text{S73})$$

$$h_f^{(\mathbf{a}_2 - \mathbf{a}_1)}(\mathbf{k}) = +2t_f \gamma_3 \tau_y. \quad (\text{S74})$$

At the half-filled saddle $h = 0$, their sum is

$$h_f(\mathbf{k}) = -2t_f \begin{pmatrix} \gamma_1 & \gamma_2 + i\gamma_3 \\ \gamma_2 - i\gamma_3 & -\gamma_1 \end{pmatrix} = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\tau} \quad (\text{S75})$$

$$\mathbf{d}(\mathbf{k}) = -2t_f (\gamma_2, -\gamma_3, \gamma_1)^T. \quad (\text{S76})$$

Here the Pauli matrices $\boldsymbol{\tau}$ act on the two magnetic sublattices, not on the physical $SU(2)$ spin. The two magnetic bands are

$$\varepsilon_{\pm}(\mathbf{k}) = \pm d(\mathbf{k}) = \pm 2t_f \sqrt{\cos^2 k_1 + \cos^2 k_2 + \cos^2(k_2 - k_1)}. \quad (\text{S77})$$

The matrix form of $h_f(\mathbf{k})$ is gauge dependent, whereas this

dispersion is not. Indeed, a lattice gauge transformation

$$f_{\mathbf{r}\sigma} \rightarrow e^{i\chi_{\mathbf{r}}} f_{\mathbf{r}\sigma}, \quad \bar{a}_{\mathbf{r}\mathbf{r}'} \rightarrow \bar{a}_{\mathbf{r}\mathbf{r}'} + \chi_{\mathbf{r}} - \chi_{\mathbf{r}'} \quad (\text{S78})$$

changes the magnetic-sublattice representation of the Bloch Hamiltonian by a unitary basis transformation, possibly together with a relabeling or folding of magnetic momentum. It therefore cannot change the eigenvalues or the gap. The Chern number and the response kernels are likewise gauge invariant when the states and current vertices are transformed consistently, although the displayed matrix entries and the components of $\mathbf{d}(\mathbf{k})$ generally change. The square root ranges from $\sqrt{3}/2$ to $\sqrt{3}$, so the direct spinon band gap is $\Delta_s = \min_{\mathbf{k}} [\varepsilon_+(\mathbf{k}) - \varepsilon_-(\mathbf{k})] = 2\sqrt{3}t_f$. At one spinon per site there are two spinons per magnetic unit cell. Consequently, for each of the two degenerate $SU(2)$ spin flavors the lower magnetic band is completely filled and the upper band is empty. The occupied band has $|C_s| = 1$ per spin flavor; we choose the orientation convention $C_s = +1$, giving total $\mathcal{C}_f = N_\sigma C_s = 2$.

Thus this Hofstadter spinon problem is still a two-band $\mathbf{d} \cdot \boldsymbol{\tau}$ model for each spin flavor, and the general response derivation below applies without change. Note that all momentum integrals run over the magnetic Brillouin zone (denoted simply by BZ below). Spatial derivatives ∂_{k_i} continue to refer to the Cartesian components $i = x, y$. In this section g_{ij} and Ω_{ij} denote the quantum metric and Berry curvature of the occupied lower Hofstadter band; we suppress the band label “−” for brevity. Since h_f is independent of the physical spin label, tracing over the $N_\sigma = 2$ degenerate copies produces the overall factor N_σ used below.

SPINON RESPONSE KERNEL FROM THE IMAGINARY-TIME PATH INTEGRAL

Quadratic action for the internal gauge field

The Peierls coupling of the spinons to a spatially uniform internal gauge field is implemented by

$$h_f(\mathbf{k}) \longrightarrow h_f(\mathbf{k} + \mathbf{a}). \quad (\text{S79})$$

Expanding to second order gives

$$h_f(\mathbf{k} + \mathbf{a}) = h_f(\mathbf{k}) + a_i \mathcal{J}_i(\mathbf{k}) + \frac{1}{2} a_i a_j \tau_{ij}(\mathbf{k}) + O(a^3), \quad (\text{S80})$$

where

$$\mathcal{J}_i(\mathbf{k}) = \partial_{k_i} h_f(\mathbf{k}), \quad \tau_{ij}(\mathbf{k}) = \partial_{k_i} \partial_{k_j} h_f(\mathbf{k}). \quad (\text{S81})$$

Here $\mathcal{J}_i$ is the paramagnetic current vertex and τ_{ij} is the diamagnetic vertex, or contact vertex.

The imaginary-time spinon action is

$$S_f[\bar{f}, f, a] = \int_0^\beta d\tau \sum_{\mathbf{k}, \sigma} \bar{f}_{\mathbf{k}\sigma} [\partial_\tau + h_f(\mathbf{k} + \mathbf{a})] f_{\mathbf{k}\sigma}. \quad (\text{S82})$$

We Fourier transform the fermion and gauge fields, with $\partial_\tau \rightarrow -i\omega_n$, according to

$$f(\tau) \propto \sum_{i\omega_n} e^{-i\omega_n \tau} f(i\omega_n), \quad a_i(\tau) \propto \sum_{i\Omega_m} e^{-i\Omega_m \tau} a_i(i\Omega_m), \quad (\text{S83})$$

where the fermionic and bosonic Matsubara frequencies are $\omega_n = (2n+1)\pi T$ and $\Omega_m = 2m\pi T$, respectively. Specifically, the gauge field a carries a bosonic Matsubara frequency Ω_m , while ω_n for the spinon is fermionic. The overall Fourier normalization factors are absorbed into the frequency sums below. Using the expansion in Eq. (S80), in Matsubara space we write the action as

$$S_f = \bar{f} (G_0^{-1} + V) f = \bar{f} (G_0^{-1} + V_1 + V_2 + \dots) f, \quad (\text{S84})$$

with the free spinon Green's function and the linear and quadratic vertices

$$G_0^{-1}(i\omega_n, \mathbf{k}) = -i\omega_n + h_f(\mathbf{k}), \\ V_1 = a_i \mathcal{J}_i, \quad V_2 = \frac{1}{2} a_i a_j \tau_{ij}. \quad (\text{S85})$$

In the absence of the gauge perturbation, the free Green's function is diagonal in frequency, momentum, and physical spin:

$$[G_0]_{(i\omega_n, \mathbf{k}, \sigma), (i\omega'_n, \mathbf{k}', \sigma')} = \delta_{\omega_n, \omega'_n} \delta_{\mathbf{k}, \mathbf{k}'} \delta_{\sigma, \sigma'} G_0(i\omega_n, \mathbf{k}), \quad (\text{S86})$$

where $G_0(i\omega_n, \mathbf{k})$ is now a two-by-two matrix in the orbital/sublattice pseudospin space.

To see how the frequency indices of the linear vertex arise, consider the term in the imaginary-time action that is first order in the gauge field,

$$S_{f,1} = \int_0^\beta d\tau \sum_{\mathbf{k}, \sigma} \bar{f}_{\mathbf{k}\sigma}(\tau) a_i(\tau) \mathcal{J}_i(\mathbf{k}) f_{\mathbf{k}\sigma}(\tau). \quad (\text{S87})$$

Substituting the Fourier expansions in Eq. (S83), and suppressing the overall Fourier normalization factors, gives

$$S_{f,1} \propto \sum_{n', n, m} \sum_{\mathbf{k}, \sigma} \bar{f}_{\mathbf{k}\sigma}(i\omega_{n'}) a_i(i\Omega_m) \mathcal{J}_i(\mathbf{k}) f_{\mathbf{k}\sigma}(i\omega_n) \\ \times \int_0^\beta d\tau e^{i(\omega_{n'} - \omega_n - \Omega_m)\tau}. \quad (\text{S88})$$

The imaginary-time integral imposes frequency conservation,

$$\int_0^\beta d\tau e^{i(\omega_{n'} - \omega_n - \Omega_m)\tau} = \beta \delta_{\omega_{n'}, \omega_n + \Omega_m}. \quad (\text{S89})$$

Thus a gauge-field mode with bosonic Matsubara frequency $i\Omega_m$ connects an incoming spinon state at $i\omega_n$ to an outgoing state at $i\omega_n + i\Omega_m$. In the uniform limit, the corresponding nonzero matrix element of V_1 is therefore

$$[V_1]_{i\omega_n + i\Omega_m, i\omega_n} = a_i(i\Omega_m) \mathcal{J}_i(\mathbf{k}), \quad (\text{S90})$$

where the momentum $\mathbf{k}$, spin σ , and orbital indices α are implicit (omitted for brevity). Its complete form reads

$$[V_1]_{(i\omega'_n, \mathbf{k}, \alpha, \sigma), (i\omega_n, \mathbf{k}, \beta, \sigma')} \\ = \delta_{\sigma\sigma'} \sum_{i\Omega_m} a_i(i\Omega_m) [\mathcal{J}_i(\mathbf{k})]_{\alpha\beta} \delta_{\omega'_n, \omega_n + \Omega_m}, \quad (\text{S91})$$

Thus, V_1 changes the spinon frequency from $i\omega_n$ to $i\omega_n + i\Omega_m$. Similarly, the quadratic vertex V_2 carries two bosonic frequencies from the two gauge fields. Its nonzero matrix element is

$$[V_2]_{i\omega_n + i\Omega_1 + i\Omega_2, i\omega_n} = \frac{1}{2} a_i(i\Omega_1) a_j(i\Omega_2) \tau_{ij}(\mathbf{k}). \quad (\text{S92})$$

Again, the momentum, spin, and orbital indices are omitted for brevity. Its complete form reads

$$[V_2]_{(i\omega'_n, \mathbf{k}, \alpha, \sigma), (i\omega_n, \mathbf{k}, \beta, \sigma')} \\ = \frac{1}{2} \delta_{\sigma\sigma'} \sum_{i\Omega_1, i\Omega_2} a_i(i\Omega_1) a_j(i\Omega_2) [\tau_{ij}(\mathbf{k})]_{\alpha\beta} \delta_{\omega'_n, \omega_n + \Omega_1 + \Omega_2}. \quad (\text{S93})$$

The effective action is further defined by the Grassmann functional integral

$$e^{-S_f^{\text{eff}}[a]} = \int \mathcal{D}\bar{f} \mathcal{D}f e^{-S_f[\bar{f}, f, a]}. \quad (\text{S94})$$

Using $\int \mathcal{D}\bar{f} \mathcal{D}f e^{-\bar{f} M f} = \det M$ and $\ln \det M = \text{Tr} \ln M$, we find

$$S_f^{\text{eff}}[a] = -\text{Tr} \ln (G_0^{-1} + V_1 + V_2 + \dots) \\ = \text{const.} - \text{Tr}(G_0 V_2) + \frac{1}{2} \text{Tr}(G_0 V_1 G_0 V_1) + O(a^3). \quad (\text{S95})$$

The two quadratic contributions in Eq. (S95) are represented diagrammatically in Fig. S2. Here $\text{Tr} := \sum_{i\omega_n} \sum_{\mathbf{k}} \sum_{\sigma} \text{tr}_{\alpha}$ includes all compound indices and frequency-momentum sums, whereas tr below acts only in the two-component orbital space. The expansion follows from

$$-\text{Tr} \ln(1 + G_0 V) = -\text{Tr}(G_0 V) + \frac{1}{2} \text{Tr}(G_0 V G_0 V) + \dots \quad (\text{S96})$$

The term linear in a_i vanishes because we expand about a saddle point with no equilibrium transport current. Since G_0 is diagonal in fermionic Matsubara frequency, we have

$$\text{Tr}(G_0 V_1 G_0 V_1) = \sum_{\omega_0, \omega_2} G_0(i\omega_0) [V_1]_{\omega_0, \omega_2} G_0(i\omega_2) [V_1]_{\omega_2, \omega_0}. \quad (\text{S97})$$

We define the quadratic kernel by

$$S_f^{(2)}[a] = \frac{1}{2} \sum_{i\Omega_m} a_i(-i\Omega_m) \Pi_f^{ij}(i\Omega_m) a_j(i\Omega_m). \quad (\text{S98})$$

For the paramagnetic contribution $\text{Tr}(G_0 V_1 G_0 V_1)$, the coeffi-

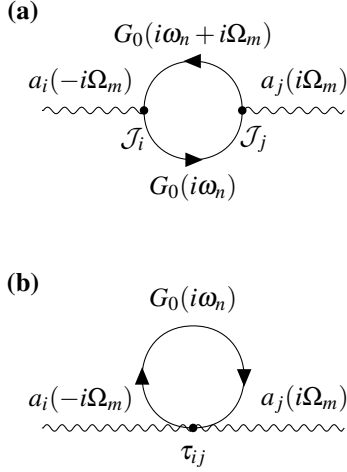


FIG. S2. Diagrammatic representation of the quadratic spinon response. (a) Paramagnetic current-current bubble generated by the two linear vertices in $\frac{1}{2}\text{Tr}(G_0 V_1 G_0 V_1)$. (b) Diamagnetic contact contribution generated by the quadratic vertex in $-\text{Tr}(G_0 V_2)$. Solid directed lines denote spinon propagators, and wavy lines denote external legs of the internal gauge field.

cient of $a_i(-i\Omega_m)a_j(i\Omega_m)$ is obtained by taking $\omega_0 = \omega_n$ and $\omega_2 = \omega_n + \Omega_m$. The first vertex then carries frequency $-i\Omega_m$ and contributes $a_i(-i\Omega_m)\mathcal{J}_i$, while the second vertex carries frequency $+i\Omega_m$ and contributes $a_j(i\Omega_m)\mathcal{J}_j$. For the diamagnetic term $-\text{Tr}(G_0 V_2)$, since V_2 already contains two powers of the gauge field, the trace contains only one spinon Green's function. Frequency conservation requires the two external gauge fields to carry opposite frequencies. This is equivalent to the standard Feynman diagram of the paramagnetic bubble and diamagnetic contact terms, as shown in Fig. S2.

For the bubble, one chooses an initial fermion frequency $i\omega_n$. One linear vertex carries $+i\Omega_m$ and takes the fermion to $i\omega_n + i\Omega_m$; the second carries $-i\Omega_m$ and closes the trace. Hence the coefficient of $a_i(-i\Omega_m)a_j(i\Omega_m)$ is

$$T \sum_{i\omega_n} \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \text{tr} [G_0(i\omega_n, \mathbf{k}) \mathcal{J}_i(\mathbf{k}) \times G_0(i\omega_n + i\Omega_m, \mathbf{k}) \mathcal{J}_j(\mathbf{k})]. \quad (\text{S99})$$

For the diamagnetic contact term, frequency conservation instead imposes $i\Omega_1 + i\Omega_2 = 0$ inside the single quadratic vertex. Comparing the resulting two terms with Eq. (S98) gives the paramagnetic bubble and the frequency-independent con-

tact term as

$$\Pi_{f,\text{para}}^{ij}(i\Omega_m) = N_\sigma T \sum_{i\omega_n} \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \text{tr} [G_0(i\omega_n, \mathbf{k}) \mathcal{J}_i(\mathbf{k}) \times G_0(i\omega_n + i\Omega_m, \mathbf{k}) \mathcal{J}_j(\mathbf{k})], \quad (\text{S100})$$

$$\Pi_{f,\text{dia}}^{ij} = -N_\sigma T \sum_{i\omega_n} \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \text{tr} [G_0(i\omega_n, \mathbf{k}) \tau_{ij}(\mathbf{k})]. \quad (\text{S101})$$

The trace in these expressions is over the two-component orbital space.

Matsubara sum and interband response

Using the band decomposition

$$G_0(i\omega_n, \mathbf{k}) = \sum_{s=\pm} \frac{|u_{s\mathbf{k}}\rangle \langle u_{s\mathbf{k}}|}{-i\omega_n + \epsilon_s(\mathbf{k})}, \quad (\text{S102})$$

the contact term can now be evaluated directly. Defining $\tau_{ij}^{ss} = \langle u_s | \tau_{ij} | u_s \rangle$ and using

$$T \sum_{i\omega_n} \frac{1}{-i\omega_n + \epsilon_s} = -n_F(\epsilon_s), \quad (\text{S103})$$

Eq. (S101) becomes

$$\Pi_{f,\text{dia}}^{ij} = N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \sum_{s=\pm} n_F(\epsilon_s) \tau_{ij}^{ss}(\mathbf{k}). \quad (\text{S104})$$

For the paramagnetic bubble we use the standard Matsubara sum

$$T \sum_{i\omega_n} \frac{1}{-i\omega_n + \epsilon_s} \frac{1}{-i(\omega_n + \Omega_m) + \epsilon_{s'}} = \frac{n_F(\epsilon_s) - n_F(\epsilon_{s'})}{i\Omega_m + \epsilon_s - \epsilon_{s'}}, \quad (\text{S105})$$

we obtain

$$\Pi_{f,\text{para}}^{ij}(i\Omega_m) = N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \sum_{s,s'=\pm} \frac{n_F(\epsilon_s) - n_F(\epsilon_{s'})}{i\Omega_m + \epsilon_s - \epsilon_{s'}} \mathcal{J}_i^{ss'} \mathcal{J}_j^{s's}, \quad (\text{S106})$$

where $\mathcal{J}_i^{ss'} = \langle u_s | \mathcal{J}_i | u_{s'} \rangle$. At zero temperature $n_F(\epsilon_-) = 1$ and $n_F(\epsilon_+) = 0$. The intraband terms therefore vanish, and the remaining interband bubble is

$$\Pi_{f,\text{para}}^{ij}(i\Omega_m) = N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \times \left[\frac{M_{ij}(\mathbf{k})}{i\Omega_m - 2d(\mathbf{k})} - \frac{M_{ji}(\mathbf{k})}{i\Omega_m + 2d(\mathbf{k})} \right], \quad (\text{S107})$$

with

$$M_{ij}(\mathbf{k}) = \langle u_- | \mathcal{J}_i | u_+ \rangle \langle u_+ | \mathcal{J}_j | u_- \rangle. \quad (\text{S108})$$

The first term in Eq. (S107) is the $s = -, s' = +$ contribution. It carries the occupation difference $n_F(\epsilon_-) - n_F(\epsilon_+) = 1$ and the excitation energy $\epsilon_+ - \epsilon_- = 2d$. The second is the $s = +, s' = -$ contribution; its occupation difference is -1 , which accounts for the explicit minus sign and gives the negative-frequency counterpart required by the current commutator.

The relation to the QGT follows directly by differentiating the band eigenvalue equation. For two different bands,

$$\langle u_{s'} | \partial_{k_i} h_f | u_s \rangle = (\epsilon_s - \epsilon_{s'}) \langle u_{s'} | \partial_{k_i} u_s \rangle. \quad (\text{S109})$$

In particular,

$$\begin{aligned} \langle u_+ | \mathcal{J}_i | u_- \rangle &= -2d \langle u_+ | \partial_{k_i} u_- \rangle, \\ \langle u_- | \mathcal{J}_i | u_+ \rangle &= -2d \langle \partial_{k_i} u_- | u_+ \rangle, \end{aligned} \quad (\text{S110})$$

where the second line also uses the derivative of $\langle u_- | u_+ \rangle = 0$. Therefore

$$M_{ij}(\mathbf{k}) = [2d(\mathbf{k})]^2 \langle \partial_{k_i} u_- | u_+ \rangle \langle u_+ | \partial_{k_j} u_- \rangle. \quad (\text{S111})$$

Since in a two-band model $1 - |u_- \rangle \langle u_-| = |u_+ \rangle \langle u_+|$, the last two matrix elements are precisely the occupied-band QGT. Hence

$$M_{ij}(\mathbf{k}) = [2d(\mathbf{k})]^2 Q_{ij}(\mathbf{k}), \quad Q_{ij} = g_{ij} - \frac{i}{2} \Omega_{ij}. \quad (\text{S112})$$

Gauge-invariant kernel and analytic continuation

We now analytically continue $i\Omega_m \rightarrow z$, where $z = \omega + i0^+$. A finite numerical broadening is included by taking $z = \omega + i\eta_s$. To display the metric and curvature contributions separately, define

$$S_{ij} = \frac{M_{ij} + M_{ji}}{2}, \quad A_{ij} = \frac{M_{ij} - M_{ji}}{2}. \quad (\text{S113})$$

Since $Q_{ji} = Q_{ij}^* = g_{ij} + i\Omega_{ij}/2$, Eq. (S112) gives

$$S_{ij} = 4d^2 g_{ij}, \quad A_{ij} = -2id^2 \Omega_{ij}. \quad (\text{S114})$$

The analytically continued paramagnetic bubble can then be rearranged as

$$\begin{aligned} \Pi_{f,\text{para}}^{ij}(z) &= N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \left\{ S_{ij} \left[\frac{1}{z-2d} - \frac{1}{z+2d} \right] \right. \\ &\quad \left. + A_{ij} \left[\frac{1}{z-2d} + \frac{1}{z+2d} \right] \right\}. \end{aligned} \quad (\text{S115})$$

Here and in the next few equations the momentum arguments of d , S_{ij} , and A_{ij} are implicit. Using

$$\begin{aligned} \frac{1}{z-2d} - \frac{1}{z+2d} &= \frac{4d}{z^2 - (2d)^2}, \\ \frac{1}{z-2d} + \frac{1}{z+2d} &= \frac{2z}{z^2 - (2d)^2}, \end{aligned} \quad (\text{S116})$$

we obtain

$$\Pi_{f,\text{para},S}^{ij}(z) = N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \frac{16d^3(\mathbf{k})}{z^2 - [2d(\mathbf{k})]^2} g_{ij}(\mathbf{k}), \quad (\text{S117})$$

$$\Pi_{f,\text{para},A}^{ij}(z) = -iN_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \frac{4zd^2(\mathbf{k})}{z^2 - [2d(\mathbf{k})]^2} \Omega_{ij}(\mathbf{k}). \quad (\text{S118})$$

The paramagnetic bubble alone is not gauge invariant: its symmetric part is nonzero for a static, uniform vector potential. The contact term in Eq. (S101) cancels this static response (f-sum rule). Equivalently, the full symmetric kernel can be written as the static subtraction

$$\Pi_{f,S}^{ij}(z) = \Pi_{f,\text{para},S}^{ij}(z) - \Pi_{f,\text{para},S}^{ij}(0). \quad (\text{S119})$$

This identity can be verified explicitly from the band expression for the contact term. At zero temperature, Eq. (S104) contains only τ_{ij}^{--} . Differentiating the lower-band energy twice gives the effective-mass identity

$$\begin{aligned} \partial_{k_i} \partial_{k_j} \epsilon_- &= \tau_{ij}^{--} + \frac{M_{ij} + M_{ji}}{\epsilon_- - \epsilon_+} \\ &= \tau_{ij}^{--} - 4d g_{ij}, \end{aligned} \quad (\text{S120})$$

where Eqs. (S112) and (S114) were used in the second line. Therefore

$$\begin{aligned} \Pi_{f,\text{dia}}^{ij} &= N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \left(\partial_{k_i} \partial_{k_j} \epsilon_- + 4d g_{ij} \right) \\ &= 4N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} d g_{ij}. \end{aligned} \quad (\text{S121})$$

The total derivative integrates to zero over the periodic Brillouin zone. On the other hand, Eq. (S117) gives

$$\Pi_{f,\text{para},S}^{ij}(0) = -4N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} d g_{ij}. \quad (\text{S122})$$

Equations (S121) and (S122) establish the static subtraction, or the corresponding f -sum rule, without leaving the contact term implicit. The antisymmetric bubble already vanishes at $z = 0$, and the contact vertex is symmetric in i, j , so the Hall part is unchanged.

Using

$$\frac{16d^3}{z^2 - (2d)^2} + 4d = \frac{4dz^2}{z^2 - (2d)^2}, \quad (\text{S123})$$

we arrive at the full retarded spinon kernel

$$\Pi_{f,S}^{ij}(z) = N_\sigma \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \frac{4d(\mathbf{k})z^2}{z^2 - [2d(\mathbf{k})]^2} g_{ij}(\mathbf{k}), \quad (\text{S124})$$

$$\Pi_{f,A}^{ij}(z) = -iN_\sigma \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \frac{4zd^2(\mathbf{k})}{z^2 - [2d(\mathbf{k})]^2} \Omega_{ij}(\mathbf{k}). \quad (\text{S125})$$

For the C_3 -symmetric triangular-lattice Hofstadter model considered here,

$$\Pi_f^{ij} = \Pi_f^L \delta^{ij} + \Pi_f^H \varepsilon^{ij}, \quad \varepsilon^{xy} = 1, \quad (\text{S126})$$

and C_3 symmetry gives $\Pi_{f,S}^{xx} = \Pi_{f,S}^{yy}$. Consequently,

$$\Pi_f^L(z) = N_\sigma \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \frac{4d(\mathbf{k})z^2}{z^2 - [2d(\mathbf{k})]^2} g_{xx}(\mathbf{k}), \quad (\text{S127})$$

$$\Pi_f^H(z) = -iN_\sigma \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \frac{4zd^2(\mathbf{k})}{z^2 - [2d(\mathbf{k})]^2} \Omega_{xy}(\mathbf{k}). \quad (\text{S128})$$

At nonzero Zeeman splitting, the corresponding band-sum derivation replaces the factor N_σ in these equations by the momentum-dependent occupation sum

$$N_\sigma \longrightarrow \sum_{\sigma=\uparrow,\downarrow} [n_F(E_{-\sigma}(\mathbf{k})) - n_F(E_{+\sigma}(\mathbf{k}))],$$

but leaves every occurrence of $d(\mathbf{k})$ unchanged. At zero temperature and below the band-reoccupation threshold, the lower band of each flavor is filled and the upper band is empty, so the bracket equals one for each σ and the sum reduces exactly to $N_\sigma = 2$. Thus the extra gap- and frequency-dependent factors displayed above are also unchanged; at finite temperature, or after Zeeman-induced band reoccupation, the occupation sum is no longer constant and the kernels do change. The static subtraction forces $\Pi_f^L \propto z^2$, while the antisymmetric response is linear in z and therefore yields a finite spinon Hall conductivity in the dc limit. In particular, below the interband threshold,

$$\begin{aligned} \Pi_f^L(\omega) &= -\chi_f^L \omega^2 + O(\omega^4), \\ \chi_f^L &= N_\sigma \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \frac{g_{xx}(\mathbf{k})}{d(\mathbf{k})} > 0, \end{aligned} \quad (\text{S129})$$

in agreement with the convention adopted in the main text.

We now separate the real and imaginary parts of Eqs. (S127) and (S128). For the retarded response, we set $z = \omega + i0^+$. At positive frequency, $(\omega + i0^+)^2 - (2d)^2$ may be written as $\omega^2 - (2d)^2 + i0^+$, with the positive factor multiplying 0^+ absorbed into its definition. The distribution identity

$$\frac{1}{x + i0^+} = \mathcal{P} \frac{1}{x} - i\pi \delta(x) \quad (\text{S130})$$

therefore gives

$$\frac{1}{(\omega + i0^+)^2 - (2d)^2} = \mathcal{P} \frac{1}{\omega^2 - (2d)^2} - i\pi \delta[\omega^2 - (2d)^2]. \quad (\text{S131})$$

For $\omega > 0$ and $d > 0$, only the positive-frequency root contributes, so that

$$\delta[\omega^2 - (2d)^2] = \frac{1}{2\omega} \delta(\omega - 2d). \quad (\text{S132})$$

Using this result in Eq. (S127), and introducing $\int [d\mathbf{k}] \equiv N_\sigma \int_{\text{BZ}} d^2k / (2\pi)^2$, we obtain

$$\text{Re} \Pi_f^L(\omega) = \mathcal{P} \int [d\mathbf{k}] \frac{4d(\mathbf{k})\omega^2}{\omega^2 - [2d(\mathbf{k})]^2} g_{xx}(\mathbf{k}), \quad (\text{S133})$$

$$\begin{aligned} \text{Im} \Pi_f^L(\omega) &= -\pi \int [d\mathbf{k}] 4d(\mathbf{k})\omega^2 g_{xx}(\mathbf{k}) \delta[\omega^2 - [2d(\mathbf{k})]^2] \\ &= -\pi\omega^2 \int [d\mathbf{k}] g_{xx}(\mathbf{k}) \delta[\omega - 2d(\mathbf{k})]. \end{aligned} \quad (\text{S134})$$

In the last line, the delta function sets $d(\mathbf{k}) = \omega/2$. Combining these two parts reproduces Eq.(8) of the main text.

For the Hall kernel, the overall factor $-i$ in Eq. (S128) interchanges the dispersive and absorptive parts. We find

$$\begin{aligned} \text{Re} \Pi_f^H(\omega) &= -\pi \int [d\mathbf{k}] 4\omega d^2(\mathbf{k}) \Omega_{xy}(\mathbf{k}) \delta[\omega^2 - [2d(\mathbf{k})]^2] \\ &= -\frac{\pi\omega^2}{2} \int [d\mathbf{k}] \Omega_{xy}(\mathbf{k}) \delta[\omega - 2d(\mathbf{k})], \end{aligned} \quad (\text{S135})$$

$$\text{Im} \Pi_f^H(\omega) = -\mathcal{P} \int [d\mathbf{k}] \frac{4\omega d^2(\mathbf{k}) \Omega_{xy}(\mathbf{k})}{\omega^2 - [2d(\mathbf{k})]^2}. \quad (\text{S136})$$

Here we again used $d(\mathbf{k}) = \omega/2$ on the support of the delta function. These two contributions combine to give Eq.(9) of the main text.

CHARGON-INDEPENDENT RECONSTRUCTION FROM KRAMERS-KRONIG RELATION

Direct inversion of Eq.(5) in the main text requires the full chargon kernel. We now show that, under a different set of experimentally testable conditions, the spinon kernel can instead be reconstructed without specifying the frequency dependence of the chargon response. The essential ingredients are the absence of a chargon Hall response, the absence of chargon absorption in the spinon spectroscopy window, and causality.

We work with retarded kernels at a complex frequency z and define the inverse circular responses

$$R_\pm(z) \equiv \frac{1}{\Pi_{\text{phys}}^\pm(z)}, \quad r_\pm(z) \equiv \frac{1}{\Pi_f^\pm(z)}, \quad q_b(z) \equiv \frac{1}{\Pi_b(z)} \quad (\text{S137})$$

The circular Ioffe-Larkin rule then becomes additive,

$$R_\pm(z) = q_b(z) + r_\pm(z), \quad (\text{S138})$$

where the same scalar q_b enters both circular channels because $\Pi_b^H = 0$. It is useful to introduce the longitudinal and Hall components of the inverse spinon kernel,

$$r_f^L \equiv \frac{r_+ + r_-}{2} = \frac{\Pi_f^L}{(\Pi_f^L)^2 + (\Pi_f^H)^2}, \quad (\text{S139a})$$

$$r_f^H \equiv \frac{r_+ - r_-}{2i} = -\frac{\Pi_f^H}{(\Pi_f^L)^2 + (\Pi_f^H)^2}. \quad (\text{S139b})$$

Likewise, $R_{\text{phys}}^L = (R_+ + R_-)/2$ and $R_{\text{phys}}^H = (R_+ - R_-)/(2i)$. Note that chargin kernel does not have a Hall component, so $q_b^\pm = q_b$. Then, from Equation (S138) we know that

$$R_{\text{phys}}^H = r_f^H, \quad R_{\text{phys}}^L = q_b + r_f^L. \quad (\text{S140})$$

Thus, the difference of the two measured inverse circular responses removes the chargin contribution exactly. Equivalently, the Hall part of the inverse physical response is purely the Hall part of the inverse spinon kernel. On the other hand, the longitudinal part of the inverse physical response is the sum of the chargin and spinon contributions, but below the onset of the chargin continuum, denoted by ω_b^{th} , the clean chargin response is reactive. Away from a zero of Π_b , both $\Pi_b(\omega)$ and $q_b(\omega)$ are then real. For $0 < \omega < \omega_b^{\text{th}}$, Eq. (S140) consequently implies

$$r_f^H(\omega) = \frac{1}{2i} \left[\frac{1}{\Pi_{\text{phys}}^+(\omega)} - \frac{1}{\Pi_{\text{phys}}^-(\omega)} \right], \quad (\text{S141a})$$

$$\text{Im } r_f^L(\omega) = \text{Im } \frac{1}{2} \left[\frac{1}{\Pi_{\text{phys}}^+(\omega)} + \frac{1}{\Pi_{\text{phys}}^-(\omega)} \right]. \quad (\text{S141b})$$

These quantities can be formed directly from the measured circular conductivities using $\Pi_{\text{phys}}^\pm = -i\omega\sigma_{\text{phys}}^\pm$, or $R_\pm = i/(\omega\sigma_{\text{phys}}^\pm)$.

The only part that remains unknown is $\text{Re } r_f^L$. This missing part is fixed by causality up to one real subtraction constant. Additional care is required since the inverse of a causal response is not automatically causal. Besides analyticity of $\Pi_f^\pm(z)$, we require

$$\Pi_f^+(z)\Pi_f^-(z) = (\Pi_f^L)^2 + (\Pi_f^H)^2 \neq 0, \quad \text{Im } z > 0, \quad (\text{S142})$$

such that $r_f^L(z)$ is analytic in the upper half-plane. Any isolated zero would instead produce a pole whose residue must be included explicitly in the dispersion relation. The zero-frequency behavior of a Chern insulator does not by itself invalidate Eq. (S142). Indeed, $\Pi_f^\pm$ vanish linearly with opposite signs as $z \rightarrow 0$; their inverse poles cancel in $(r_+ + r_-)/2$, leaving $r_f^L(0)$ finite. And we have checked that $r_f^L(z)$ is indeed analytic in the upper half-plane.

We then have the following Kramers–Kronig relation to ob-

tain the $\text{Re } r_f^L$:

$$\text{Re } r_f^L(\omega) = r_{f,\infty}^L + \frac{2}{\pi} \mathcal{P} \int_0^\infty d\omega' \frac{\omega' \text{Im } r_f^L(\omega')}{\omega'^2 - \omega^2}, \quad (\text{S143})$$

where $r_{f,\infty}^L \equiv \lim_{|z| \rightarrow \infty} r_f^L(z)$ is real. For numerical evaluation with an upper cutoff Ω_{cut} , the principal-value singularity at $\omega' = \omega$ can be removed analytically. Define $f_r(\omega') \equiv \omega' \text{Im } r_f^L(\omega')$. For $0 < \omega < \Omega_{\text{cut}}$, subtracting and adding $f_r(\omega)$ gives the exact identity

$$\mathcal{P} \int_0^{\Omega_{\text{cut}}} \frac{f_r(\omega')}{\omega'^2 - \omega^2} d\omega' = \int_0^{\Omega_{\text{cut}}} \frac{f_r(\omega') - f_r(\omega)}{\omega'^2 - \omega^2} d\omega' + \frac{f_r(\omega)}{2\omega} \ln \left| \frac{\Omega_{\text{cut}} - \omega}{\Omega_{\text{cut}} + \omega} \right|. \quad (\text{S144})$$

If f_r is continuously differentiable near ω , as in our finite-broadening calculation, the integrand on the right has the finite limit

$$\lim_{\omega' \rightarrow \omega} \frac{f_r(\omega') - f_r(\omega)}{\omega'^2 - \omega^2} = \frac{f_r'(\omega)}{2\omega}. \quad (\text{S145})$$

The logarithmic term is the analytically integrated principal value of the subtracted part and accounts for the cancellation of the two-sided singularity. In our numerical implementation, we use the above limit at the coincident grid point, evaluate the regular integral with the trapezoidal rule, and add back the logarithmic term. The endpoint $\omega = \Omega_{\text{cut}}$ is excluded, and the cutoff is chosen above the frequency range of interest; any omitted spectral tail remains a separate truncation error.

Basing on the mean-field kernel in Eqs. (S127) and (S128), we know

$$r_{f,\infty}^L = \left[N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} 4d(\mathbf{k}) g_{xx}(\mathbf{k}) \right]^{-1}. \quad (\text{S146})$$

In practice, “infinity” denotes the asymptotic regime of the low-energy spinon theory—frequencies above the spinon bandwidth but still below the scale at which the parton description itself ceases to apply. Note that at $z \rightarrow \infty$, $\Pi_f^H(z)$ approaches to zero while $\Pi_f^L(z)$ approaches the quantity inside the square brackets of above equation. Thus, the inverse constant $(r_{f,\infty}^L)^{-1}$ is related to the quantum metric as

$$D_f \equiv \lim_{|z| \rightarrow \infty} \Pi_f^L(z) = (r_{f,\infty}^L)^{-1} = N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} 4d(\mathbf{k}) g_{xx}(\mathbf{k}). \quad (\text{S147})$$

For the triangular Hofstadter ansatz, D_f can be determined by the measured spinon interband threshold. Note that the spinon eigenvectors and hence g_{xx} are independent of t_f , so we can write

$$D_f = t_f \mathcal{D}_\Delta, \quad \mathcal{D}_\Delta \equiv N_\sigma \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} 4\tilde{d}(\mathbf{k}) g_{xx}(\mathbf{k}). \quad (\text{S148})$$

where the spinon hopping parameter t_f is separated out from

$\tilde{d}(\mathbf{k})$ as $d(\mathbf{k}) = t_f \tilde{d}(\mathbf{k})$. Thus, $\mathcal{D}_\Delta$ is a dimensionless quantity that only depends on the lattice geometry instead of spinon hopping parameter. Numerical integration over the magnetic Brillouin zone gives $\mathcal{D}_\Delta = 1.38681422$ for $N_\sigma = 2$. The spinon hopping t_f can then be inferred from the measured interband threshold, $\Delta_s = 2\sqrt{3}t_f$. Consequently, the subtraction constant is calibrated as

$$r_{f,\infty}^L = \frac{1}{D_f} = \frac{2.49788441}{\Delta_s}. \quad (\text{S149})$$

Although it relies on the spinon ansatz on triangular-lattice, this calibration is independent of the chargon model.

An isolated spinon is gauge charged, so a local spectroscopic probe does not directly measure $\Delta_f = \Delta_s/2$. What is measured is the gauge-invariant particle-hole threshold Δ_s . Experimentally, the two complex circular conductivities determine

$$Y_L(\omega) \equiv \text{Im} \frac{1}{2} \left[\frac{i}{\omega \sigma_{\text{phys}}^+(\omega)} + \frac{i}{\omega \sigma_{\text{phys}}^-(\omega)} \right] = \text{Im} r_f^L(\omega) \quad (\text{S150})$$

below the chargon absorption threshold. In the clean limit, Y_L has no absorptive support below Δ_s , and its first interband edge fixes Δ_s . Thus, Δ_s is obtained by fitting the broadened leading edge (first dashed line in Fig. S3). The present dispersion further predicts support over $\Delta_s \leq \omega \leq 2\Delta_s$, so the upper edge (second dashed line in Fig. S3) also provides an internal check of the assumed band shape.

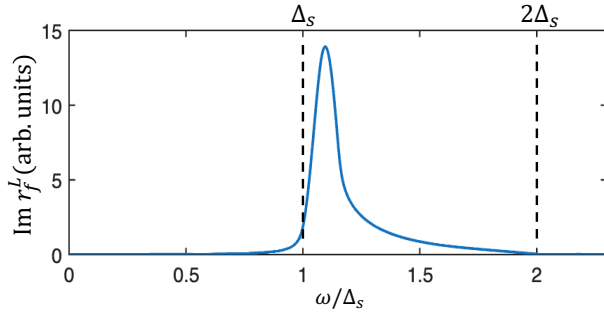


FIG. S3. Distribution of $\text{Im} r_f^L$ (arbitrary units) over the spinon interband continuum with $U/t = 15$ and $\eta_s/t_f = 0.05$. The dashed vertical lines indicate the spinon interband threshold Δ_s and the upper edge $2\Delta_s$.

There is also a spectral-separation requirement. Equation (S143) needs $\text{Im} r_f^L$ over its full absorptive support, whereas Eq. (S141) isolates it only where the chargon is lossless. A strictly chargon-independent closure is therefore possible when the entire spinon interband continuum relevant to the reconstruction lies below ω_b^{th} , so that $\text{Im} r_f^L$ has vanished before chargon absorption begins. If the two continua overlap, the unmeasured high-frequency part of the Kramers–Kronig integral must be supplied by an additional model.

For the present finite-band ansatz the clean spinon support ends at $2\Delta_s$. Hence, if $2\Delta_s < \Omega_{\text{cut}} < \omega_b^{\text{th}}$, the upper limit in

Eq. (S143) can be replaced by Ω_{cut} exactly: an infinite experimental scan is unnecessary. Weak high-energy tails or overlap with the chargon continuum must instead be included as a finite-window systematic uncertainty.

After completing r_f^L , the circular and Cartesian spinon kernels are obtained algebraically:

$$r_{\pm} = r_f^L \pm i r_f^H, \quad \Pi_f^{\pm} = \frac{1}{r_{\pm}}, \quad (\text{S151a})$$

$$\Pi_f^L = \frac{\Pi_f^+ + \Pi_f^-}{2}, \quad \Pi_f^H = \frac{\Pi_f^+ - \Pi_f^-}{2i}. \quad (\text{S151b})$$

Substitution into Eq.(19) in the main text finally yields the quantum-metric and Berry-curvature spectral densities. This construction does not require a microscopic frequency-dependent chargon model. It does require a vanishing chargon Hall channel, negligible chargon absorption over the spinon support, an upper-half-plane zero-free inverse spinon kernel (or explicit pole terms), and the scalar normalization fixed from the measured interband threshold.

Alternative method: first moment of the dynamical spin structure factor

As an independent experimental cross-check for $r_{f,\infty}^L$, one may utilize the small-momentum first moment of the dynamical spin structure factor (DSSF). At zero temperature, with $S^z = (n_{\uparrow} - n_{\downarrow})/2$ and $\omega_{n0} = E_n - E_0 > 0$, we define

$$S^{zz}(\mathbf{q}, \omega) \equiv \sum_n |\langle n | S_q^z | 0 \rangle|^2 \delta(\omega - \omega_{n0}). \quad (\text{S152})$$

If S^z is conserved, its long-wavelength continuity equation is

$$\partial_t S_q^z + i q J_L^z(\mathbf{q}) = O(q^2), \quad (\text{S153})$$

where $J_L^z = \hat{\mathbf{q}} \cdot \mathbf{J}^z$. Taking a matrix element between the ground state and an excited state gives

$$\omega_{n0} \langle n | S_q^z | 0 \rangle = -q \langle n | J_L^z(\mathbf{q}) | 0 \rangle + O(q^2). \quad (\text{S154})$$

The sign depends on the Fourier-transform convention and drops out after taking the modulus squared. Introducing the positive-frequency longitudinal spin-current spectrum

$$S_{JJ}^L(\mathbf{q}, \omega) \equiv \sum_n |\langle n | J_L^z(\mathbf{q}) | 0 \rangle|^2 \delta(\omega - \omega_{n0}), \quad (\text{S155})$$

the continuity equation therefore yields

$$\lim_{q \rightarrow 0} S_{JJ}^L(\mathbf{q}, \omega) = \lim_{q \rightarrow 0} \frac{\omega^2}{q^2} S^{zz}(\mathbf{q}, \omega). \quad (\text{S156})$$

To connect this spectrum to an instantaneous response weight, we write the retarded physical spin-current kernel as

$$\Pi_{f,\text{spin}}^L(z) = \mathcal{D}_f + \chi_{JJ}^R(z), \quad (\text{S157})$$

which should be distinguished from the internal-gauge current kernel Π_f^L in Eq. (S127). Here $\mathcal{D}_f$ is the frequency-independent diamagnetic/contact term. The paramagnetic term has the Lehmann representation

$$\chi_{JJ}^R(z) = \int_0^\infty d\omega' \mathcal{S}_{JJ}^L(\mathbf{0}, \omega') \left(\frac{1}{z - \omega'} - \frac{1}{z + \omega'} \right). \quad (\text{S158})$$

The gapped spin liquid has no spin stiffness, so the static uniform kernel vanishes, $\Pi_{f,\text{spin}}^L(\mathbf{0}) = 0$. It follows that

$$\begin{aligned} \mathcal{D}_f &= -\chi_{JJ}^R(0) = 2 \int_0^\infty \frac{d\omega}{\omega} \mathcal{S}_{JJ}^L(\mathbf{0}, \omega) \\ &= 2 \lim_{q \rightarrow 0} \frac{1}{q^2} \int_0^\infty d\omega \omega S^{\text{zz}}(\mathbf{q}, \omega). \end{aligned} \quad (\text{S159})$$

Before comparing $\mathcal{D}_f$ with the internal-gauge weight D_f , we make the origin of the single-flavor quantity D_0 explicit. Removing the overall factor N_σ from Eqs. (S117) and (S121), the longitudinal kernel of one flavor with unit probe charge is

$$\Pi_0^L(z) = D_0 + \Pi_{0,\text{para}}^L(z), \quad (\text{S160})$$

$$D_0 = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} 4d(\mathbf{k}) g_{xx}(\mathbf{k}), \quad (\text{S161})$$

$$\Pi_{0,\text{para}}^L(z) = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \frac{16d^3(\mathbf{k})}{z^2 - [2d(\mathbf{k})]^2} g_{xx}(\mathbf{k}). \quad (\text{S162})$$

The last line decays as z^{-2} for $|z| \rightarrow \infty$, whereas the contact term is frequency independent. Therefore

$$\lim_{|z| \rightarrow \infty} \Pi_0^L(z) = D_0. \quad (\text{S163})$$

Equivalently, this follows by setting $N_\sigma = 1$ in Eq. (S127) before taking the high-frequency limit. Thus D_0 is not an additional phenomenological parameter: it is the single-flavor diamagnetic weight derived from the same spinon Hamiltonian.

It remains to account for the probe charges carried by the two flavors. Both flavors carry unit internal gauge charge, whereas their physical S^z charges are $+1/2$ and $-1/2$. Since the two contributions are equal and cross-flavor correlators vanish,

$$D_f = (1^2 + 1^2)D_0 = 2D_0, \quad (\text{S164a})$$

$$D_f = \left[\left(\frac{1}{2} \right)^2 + \left(-\frac{1}{2} \right)^2 \right] D_0 = \frac{D_0}{2} = \frac{D_f}{4}. \quad (\text{S164b})$$

Consequently,

$$D_f = 8 \lim_{q \rightarrow 0} \frac{1}{q^2} \int_0^\infty d\omega \omega S^{\text{zz}}(\mathbf{q}, \omega). \quad (\text{S165})$$

The right-hand side can in principle be determined from absolutely normalized, polarization-resolved inelastic neutron scattering. This test requires the full magnetic continuum, a controlled small- q extrapolation, and an approximately con-

served spin component. The continuity equation is general under these conditions, whereas $D_f = 4\mathcal{D}_f$ additionally assumes the present two-flavor mean-field saddle and can receive interaction or vertex corrections.

SELF-CONSISTENT MEAN-FIELD METHOD

We first clarify which quantities are independent inputs and which are outputs of the parton saddle. This distinction is important because the parameters appearing in the two parton Hamiltonians cannot all be varied independently if they are to represent one microscopic Hofstadter–Hubbard model.

The triangular lattice, half filling, and the number of flavors $N_\sigma = 2$ are fixed throughout the work apparently. The bare electron hopping t sets the microscopic energy unit. The orbital flux $\Phi_\Delta = \pi/2$ is imposed externally rather than obtained from a mean-field equation, and $\bar{A}_{ij}$ is a gauge representative of this fixed flux. Likewise, U/t is a tunable microscopic control parameter, not a saddle-point output. It may be chosen within the CSL interval established for the same model, preferably away from either phase boundary [34, 35]. We work at $T = 0$ for a clean exposure of the spinon quantum geometry (otherwise, there will be additional distribution factors in the optical kernels). We ignore the Zeeman splitting. A nonzero Zeeman splitting below the spin gap does not modify the spin-conserving optical kernels at zero temperature before the band occupations change, as explained in the discussion following Eq. (S16).

The gauge-invariant flux ansatz of the adopted saddle is the flux partition

$$\Phi_f = \sum_{\partial\Delta} \bar{a}_{ij} = \frac{\pi}{2}, \quad \Phi_b = \sum_{\partial\Delta} (\bar{A}_{ij} - \bar{a}_{ij}) = 0.$$

This uniform CSL branch, including its two-site spinon magnetic unit cell and topological band filling, can be adopted from the self-consistent and many-body studies[34, 35]; The bond-by-bond choices $\bar{a}_{ij} = \bar{A}_{ij}$, $\phi_{ij}^b = 0$, and $\phi_{ij}^f = -\bar{A}_{ij}$ are gauge conventions. The unbarred field a_μ represents a dynamical gauge fluctuation around this saddle. It is integrated out in the tensor Ioffe–Larkin rule and is neither an externally chosen number nor an additional mean-field fitting parameter. The weak field A_μ is the optical source used to probe the response.

The Lagrangian multiplier λ can be determined once U_s and t_b are fixed. Performing the Matsubara sum in Eq. (S46) gives

$$1 = \frac{1}{N} \sum_{\mathbf{k}} \frac{U_s}{E_b(\mathbf{k})} \coth \left[\frac{E_b(\mathbf{k})}{2T} \right]. \quad (\text{S166})$$

Hence at zero temperature the triangular-lattice equation reduces to

$$1 = \int_{\mathbf{k}} \frac{U_s}{\sqrt{2U_s[\lambda - 2t_b\gamma_\Delta(\mathbf{k})]}}, \quad \lambda > 6t_b. \quad (\text{S167})$$

This is a one-dimensional root-finding problem for λ . Its uncondensed solution is unique whenever the selected parameters lie on the Mott side of the rotor condensation transition. For $U_s \gg t_b$, using $\langle \epsilon_b \rangle_{\mathbf{k}} = 0$ and $\langle \epsilon_b^2 \rangle_{\mathbf{k}} = 6t_b^2$, one has the expansion:

$$\lambda = \frac{U_s}{2} + \frac{9t_b^2}{U_s} + O(t_b^3/U_s^2).$$

We now evaluate directly the bond-field definitions in Eqs. (S5). In the gauge of Eq. (S11), they read

$$\chi_{ij}^b = |\chi_{ij}^b|, \quad \chi_{ij}^f = |\chi_{ij}^f| e^{-i\tilde{A}_{ij}}.$$

Thus χ_{ij}^f itself is generally complex. Magnetic translations and the (projective) C_3 symmetry make the magnitudes identical on all nearest-neighbor bonds. We may therefore suppress the bond indices on the magnitudes and write $|\chi_{ij}^b| = |\chi^b|$ and $|\chi_{ij}^f| = |\chi^f|$, while retaining the bond-dependent Peierls phase explicitly. These bond fields should not be confused with the low-frequency chargin polarizability χ_b , whose subscript carries a lower parton label. From Eq. (S5) and Eq. (S8), the two hopping amplitudes satisfy

$$\frac{t_b}{t} = |\chi^f| = \frac{N_\sigma}{6} \langle \rho(\mathbf{k}) \rangle_{\text{MBZ}}, \quad (\text{S168a})$$

$$\frac{t_f}{t} = |\chi^b| = \int_{\mathbf{k}} \cos(\mathbf{k} \cdot \delta_{\mathbf{v}}) \frac{U_s}{E_b(\mathbf{k})} = \frac{1}{3} \int_{\mathbf{k}} \gamma_{\Delta}(\mathbf{k}) \frac{U_s}{E_b(\mathbf{k})}. \quad (\text{S168b})$$

Here $N_\sigma = 2$, $\rho(\mathbf{k}) = \sqrt{\cos^2 k_1 + \cos^2 k_2 + \cos^2(k_2 - k_1)}$, and $E_b(\mathbf{k}) = \sqrt{2U_s[\lambda - 2t_b\gamma_{\Delta}(\mathbf{k})]}$. The λ entering the second line is not an additional unresolved quantity: once the first line fixes t_b , it is obtained numerically from the zero-temperature constraint in Eq. (S167) above.

We now clarify how Eq. (S168a) can be derived from Eq. (S5b). Let N_m be the number of two-site magnetic unit cells, so that the number of lattice sites is $N_s = 2N_m$, and define $\langle O \rangle_{\text{MBZ}} = N_m^{-1} \sum_{\mathbf{k} \in \text{MBZ}} O(\mathbf{k})$. At half filling the lower spinon band is completely filled for each of the $N_\sigma = 2$ spin flavors. Using $\epsilon_{-}(\mathbf{k}) = -2t_f \rho(\mathbf{k})$, its ground-state energy per site is

$$\frac{E_f^{(0)}}{N_s} = \frac{N_\sigma}{2N_m} \sum_{\mathbf{k} \in \text{MBZ}} \epsilon_{-}(\mathbf{k}) = -N_\sigma t_f \langle \rho(\mathbf{k}) \rangle_{\text{MBZ}}.$$

The same expectation value evaluated from the real-space spinon Hamiltonian is

$$E_f^{(0)} = -t_f \sum_{\langle ij \rangle} \left[e^{i\tilde{A}_{ij}} \chi_{ij}^f + \text{c.c.} \right] = -2t_f |\chi^f| (3N_s)$$

Here the phase in $\chi_{ij}^f = |\chi^f| e^{-i\tilde{A}_{ij}}$ cancels the explicit Peierls phase. The factor $3N_s$ is the number of undirected nearest-neighbor bonds on a triangular lattice: the coordination number is six, and each bond is shared by two sites. Equat-

ing the momentum- and real-space expressions gives $|\chi^f| = (N_\sigma/6) \langle \rho \rangle_{\text{MBZ}}$, which is Eq. (S168a). This is also the Hellmann–Feynman evaluation of the occupied-band projector. Because the lower band is completely filled, its eigenvectors and ρ do not depend on the overall scale t_f , so this equation directly determines $t_b = t |\chi^f|$ without an initial guess for t_f .

We next derive Eq. (S168b) from Eq. (S5a). Since $b_i = X_i = e^{i\theta_i}$, the zero-flux gauge gives, for a representative bond $j = i + \delta_{\mathbf{v}}$,

$$\begin{aligned} |\chi^b| &= \langle X_i^* X_{i+\delta_{\mathbf{v}}} \rangle \\ &= \frac{T}{N_s} \sum_{\mathbf{k}, \mathbf{v}_n} e^{i\mathbf{k} \cdot \delta_{\mathbf{v}}} G_b(\mathbf{k}, i\mathbf{v}_n). \end{aligned}$$

Because $E_b(\mathbf{k}) = E_b(-\mathbf{k})$, the sine part of the exponential integrates to zero. Substituting $G_b = 2U_s/[v_n^2 + E_b^2(\mathbf{k})]$ from Eq. (S45) and using the bosonic sum

$$T \sum_{\mathbf{v}_n} \frac{1}{v_n^2 + E^2} = \frac{1}{2E} \coth\left(\frac{E}{2T}\right)$$

yields

$$|\chi^b| = \int_{\mathbf{k}} \cos(\mathbf{k} \cdot \delta_{\mathbf{v}}) \frac{U_s}{E_b(\mathbf{k})} \coth\left[\frac{E_b(\mathbf{k})}{2T}\right].$$

At $T = 0$, the hyperbolic cotangent tends to one. Moreover, C_3 symmetry makes this correlator equal for the three bond directions. Summing over $\mathbf{v} = 1, 2, 3$ and using $\gamma_{\Delta}(\mathbf{k}) = \sum_{\mathbf{v}=1}^3 \cos(\mathbf{k} \cdot \delta_{\mathbf{v}})$ then gives the two equivalent forms in Eq. (S168b).

Consequently, the actual sequence of self-consistent calculations is: solve Eq. (S167) for the multiplier λ for the soft mode constraint, evaluate the first line of Eq. (S168) to obtain t_b , and evaluate the second line of Eq. (S168) to obtain t_f . The mean-field solutions for each U/t are shown in Fig. S4.

Existing works can only be used directly for the existence and flux structure of the CSL branch and for locating a reasonable interval of U/t . Previous works [34, 35] do not tabulate a compatible pair (t_b, t_f) for our atomic-limit calibration $U_s = U/2$; in particular, results from a strong-coupling spinon-only saddle do not determine the Gaussian rotor bond in Eq. (S168). Therefore the flux branch can be adopted from the literature, whereas a quantitatively microscopic kernel at a selected U/t should use Eq. (S168) together with the already given λ constraint to recompute its two bond amplitudes in our convention.

Once U_s , t_b , and λ are known, the gap and band width of Gaussian chargin can be derived from

$$\Delta_b = \sqrt{2U_s(\lambda - 6t_b)}, \quad (\text{S169})$$

$$W_b = \sqrt{\Delta_b^2 + 18U_s t_b} - \Delta_b, \quad (\text{S170})$$

Additionally, χ_b , γ_b , and the full $\Pi_b(\omega)$ through the Brillouin-zone integrals can also be derived. Similarly, once

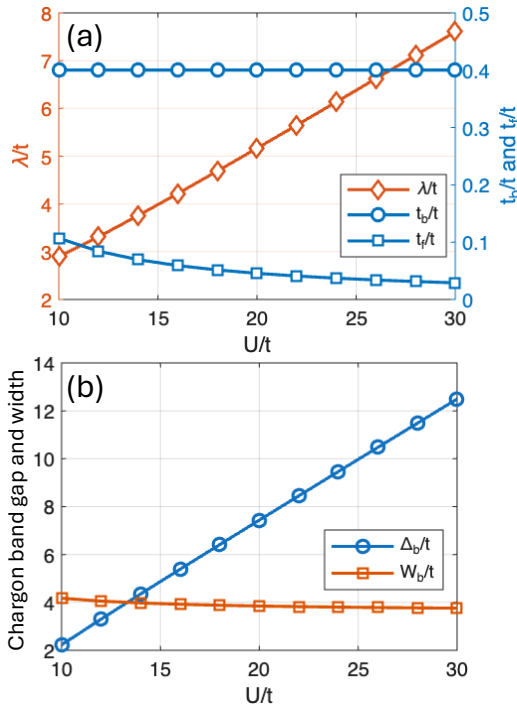


FIG. S4. (a) Mean-field solutions for λ , t_b , and t_f as a function of U/t . (b) The charged band gap and bandwidth calculated from Eq. (S170).

t_f is known, the energy scale of the spinon band and spinon kernel $\Pi_f^{ij}(\omega)$ can be determined. Note that the spinon QGT is independent of the overall scale t_f .

DERIVATION OF THE QUANTUM GEOMETRIC TENSOR

In this appendix we derive the quantum geometric tensor (QGT) for a generic two-band Hamiltonian using the projector formalism. Throughout this appendix we use the convention

$$Q_{ij} = g_{ij} - \frac{i}{2}\Omega_{ij}, \quad (\text{S171})$$

where g_{ij} is the quantum metric and Ω_{ij} is the Berry curvature.

Projector representation of the QGT

Consider a single nondegenerate Bloch band with normalized eigenstate $|u(\mathbf{k})\rangle$. The band projector is

$$P(\mathbf{k}) = |u(\mathbf{k})\rangle\langle u(\mathbf{k})|. \quad (\text{S172})$$

The quantum geometric tensor is defined as

$$Q_{ij} \equiv \langle \partial_i u | (1 - P) | \partial_j u \rangle, \quad \partial_i \equiv \partial_{k_i}. \quad (\text{S173})$$

This form is invariant under the local $U(1)$ gauge transformation $|u(\mathbf{k})\rangle \rightarrow e^{i\varphi(\mathbf{k})}|u(\mathbf{k})\rangle$, because the projector $1 - P$ removes the component of $|\partial_j u\rangle$ parallel to $|u\rangle$.

We now show that the same tensor can be written in terms of projectors. Taking the derivative of $P = |u\rangle\langle u|$, one obtains

$$\partial_i P = |\partial_i u\rangle\langle u| + |u\rangle\langle \partial_i u|. \quad (\text{S174})$$

Using the normalization condition $\langle u|u\rangle = 1$, one has

$$\langle \partial_i u | u \rangle = -\langle u | \partial_i u \rangle. \quad (\text{S175})$$

It follows that

$$(\partial_j P)|u\rangle = (1 - P)|\partial_j u\rangle, \quad (\text{S176})$$

and

$$\langle u | (\partial_i P) = \langle \partial_i u | (1 - P). \quad (\text{S177})$$

Therefore

$$\begin{aligned} \text{Tr}[P\partial_i P\partial_j P] &= \langle u | \partial_i P \partial_j P | u \rangle \\ &= \langle \partial_i u | (1 - P)^2 | \partial_j u \rangle \\ &= \langle \partial_i u | (1 - P) | \partial_j u \rangle \\ &= Q_{ij}. \end{aligned} \quad (\text{S178})$$

Similarly,

$$\text{Tr}[P\partial_j P\partial_i P] = Q_{ji}. \quad (\text{S179})$$

The antisymmetric part gives the Berry curvature. From

$$\begin{aligned} \text{Tr}[P[\partial_i P, \partial_j P]] &= \text{Tr}[P\partial_i P\partial_j P] - \text{Tr}[P\partial_j P\partial_i P] \\ &= Q_{ij} - Q_{ji}, \end{aligned} \quad (\text{S180})$$

and using the convention $Q_{ij} = g_{ij} - i\Omega_{ij}/2$, we find

$$Q_{ij} - Q_{ji} = -i\Omega_{ij}. \quad (\text{S181})$$

Thus the Berry curvature can be written as

$$\Omega_{ij} = i\text{Tr}[P[\partial_i P, \partial_j P]]. \quad (\text{S182})$$

The symmetric part gives the quantum metric. Since $Q_{ji} = Q_{ij}^*$,

$$g_{ij} = \text{Re } Q_{ij} = \frac{1}{2}(Q_{ij} + Q_{ji}). \quad (\text{S183})$$

From the definition of the QGT in Eq. (S173), one has

$$\begin{aligned} Q_{ij} + Q_{ji} &= \langle \partial_i u | \partial_j u \rangle + \langle \partial_j u | \partial_i u \rangle \\ &\quad + \langle \partial_i u | u \rangle \langle \partial_j u | u \rangle + \langle \partial_j u | u \rangle \langle \partial_i u | u \rangle. \end{aligned} \quad (\text{S184})$$

On the other hand, using the identity $\text{Tr}(|a\rangle\langle b|) = \langle a|b\rangle$, we

have

$$\begin{aligned} \text{Tr}[\partial_i P \partial_j P] &= \langle \partial_i u | \partial_j u \rangle + \langle \partial_j u | \partial_i u \rangle \\ &+ \langle \partial_i u | u \rangle \langle \partial_j u | u \rangle + \langle u | \partial_j u \rangle \langle u | \partial_i u \rangle. \end{aligned} \quad (\text{S185})$$

Note that $\langle u | \partial_j u \rangle \langle u | \partial_i u \rangle = \langle \partial_j u | u \rangle \langle \partial_i u | u \rangle$. Therefore, based on the preceding two equations, one obtains

$$g_{ij} = \frac{1}{2} \text{Tr}[\partial_i P \partial_j P]. \quad (\text{S186})$$

Hence, for the convention used here,

$$g_{ij} = \frac{1}{2} \text{Tr}[\partial_i P \partial_j P], \quad \Omega_{ij} = i \text{Tr}[P[\partial_i P, \partial_j P]]. \quad (\text{S187})$$

Application to triangular-lattice Hofstadter spinons

Now consider a generic two-band Hamiltonian

$$h(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\tau}, \quad (\text{S188})$$

where $\boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z)$ are Pauli matrices in the two-band space. We define

$$d(\mathbf{k}) = |\mathbf{d}(\mathbf{k})|, \quad \hat{\mathbf{d}}(\mathbf{k}) = \frac{\mathbf{d}(\mathbf{k})}{d(\mathbf{k})}. \quad (\text{S189})$$

The Hamiltonian can be written as

$$h(\mathbf{k}) = d(\mathbf{k}) \hat{\mathbf{d}}(\mathbf{k}) \cdot \boldsymbol{\tau}. \quad (\text{S190})$$

Since

$$(\hat{\mathbf{d}} \cdot \boldsymbol{\tau})^2 = 1, \quad (\text{S191})$$

the eigenvalues of $\hat{\mathbf{d}} \cdot \boldsymbol{\tau}$ are ± 1 , and the band energies are

$$\varepsilon_{\pm}(\mathbf{k}) = \pm d(\mathbf{k}). \quad (\text{S192})$$

The projectors onto the upper and lower bands are therefore

$$P_{\pm}(\mathbf{k}) = |u_{\pm}\rangle \langle u_{\pm}| = \frac{1}{2} [1 \pm \hat{\mathbf{d}}(\mathbf{k}) \cdot \boldsymbol{\tau}]. \quad (\text{S193})$$

For the lower band,

$$P_{-}(\mathbf{k}) = \frac{1}{2} [1 - \hat{\mathbf{d}}(\mathbf{k}) \cdot \boldsymbol{\tau}]. \quad (\text{S194})$$

One can verify that $P_{-}^2 = P_{-}$, $P_{-} |u_{-}\rangle = |u_{-}\rangle$, and $P_{-} |u_{+}\rangle = 0$. Taking a momentum derivative gives

$$\partial_i P_{-} = -\frac{1}{2} (\partial_i \hat{\mathbf{d}}) \cdot \boldsymbol{\tau}. \quad (\text{S195})$$

We first compute the quantum metric. In 2D, using

$$(\mathbf{a} \cdot \boldsymbol{\tau})(\mathbf{b} \cdot \boldsymbol{\tau}) = (\mathbf{a} \cdot \mathbf{b}) \mathbb{I} + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\tau}, \quad (\text{S196})$$

and

$$\text{Tr} \mathbb{I} = 2, \quad \text{Tr} \tau_a = 0, \quad (\text{S197})$$

one finds

$$\begin{aligned} \text{Tr}[\partial_i P_{-} \partial_j P_{-}] &= \frac{1}{4} \text{Tr}[(\partial_i \hat{\mathbf{d}} \cdot \boldsymbol{\tau})(\partial_j \hat{\mathbf{d}} \cdot \boldsymbol{\tau})] \\ &= \frac{1}{2} \partial_i \hat{\mathbf{d}} \cdot \partial_j \hat{\mathbf{d}}. \end{aligned} \quad (\text{S198})$$

Therefore

$$g_{ij}^{-} = \frac{1}{4} \partial_i \hat{\mathbf{d}} \cdot \partial_j \hat{\mathbf{d}}. \quad (\text{S199})$$

Next we compute the Berry curvature. Using the identity

$$[\mathbf{a} \cdot \boldsymbol{\tau}, \mathbf{b} \cdot \boldsymbol{\tau}] = 2i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\tau} \quad (\text{S200})$$

the commutator of the projector derivatives is

$$\begin{aligned} [\partial_i P_{-}, \partial_j P_{-}] &= \frac{1}{4} [\partial_i \hat{\mathbf{d}} \cdot \boldsymbol{\tau}, \partial_j \hat{\mathbf{d}} \cdot \boldsymbol{\tau}] \\ &= \frac{i}{2} (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}) \cdot \boldsymbol{\tau}. \end{aligned} \quad (\text{S201})$$

Using

$$P_{-} = \frac{1}{2} [1 - \hat{\mathbf{d}} \cdot \boldsymbol{\tau}], \quad (\text{S202})$$

we obtain

$$\text{Tr}[P_{-} (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}) \cdot \boldsymbol{\tau}] = -\hat{\mathbf{d}} \cdot (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}). \quad (\text{S203})$$

Thus

$$\text{Tr}[P_{-} [\partial_i P_{-}, \partial_j P_{-}]] = -\frac{i}{2} \hat{\mathbf{d}} \cdot (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}). \quad (\text{S204})$$

With the convention $\Omega_{ij} = i \text{Tr}\{P[\partial_i P, \partial_j P]\}$, this gives

$$\Omega_{ij}^{-} = \frac{1}{2} \hat{\mathbf{d}} \cdot (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}). \quad (\text{S205})$$

Combining the metric and Berry curvature, the lower-band QGT is

$$Q_{ij}^{-} = \frac{1}{4} [\partial_i \hat{\mathbf{d}} \cdot \partial_j \hat{\mathbf{d}} - i \hat{\mathbf{d}} \cdot (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}})]. \quad (\text{S206})$$

It is often useful to rewrite these expressions directly in terms of $\mathbf{d}$. Since

$$\partial_i \hat{\mathbf{d}} = \frac{\partial_i \mathbf{d}}{d} - \frac{\mathbf{d}(\mathbf{d} \cdot \partial_i \mathbf{d})}{d^3}, \quad (\text{S207})$$

the quantum metric becomes

$$g_{ij}^{-} = \frac{1}{4} \left[\frac{\partial_i \mathbf{d} \cdot \partial_j \mathbf{d}}{d^2} - \frac{(\mathbf{d} \cdot \partial_i \mathbf{d})(\mathbf{d} \cdot \partial_j \mathbf{d})}{d^4} \right]. \quad (\text{S208})$$

Similarly,

$$\hat{\mathbf{d}} \cdot (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}) = \frac{\mathbf{d} \cdot (\partial_i \mathbf{d} \times \partial_j \mathbf{d})}{d^3}, \quad (\text{S209})$$

and therefore

$$\Omega_{ij}^- = \frac{\mathbf{d} \cdot (\partial_i \mathbf{d} \times \partial_j \mathbf{d})}{2d^3}. \quad (\text{S210})$$

We now apply these expressions to the triangular-lattice Hofstadter spinon Hamiltonian in Eq. (S76). Define

$$k_1 = ak_x, \quad k_2 = \frac{a}{2}(\sqrt{3}k_y + k_x), \quad k_3 \equiv k_2 - k_1 = \frac{a}{2}(\sqrt{3}k_y - k_x),$$

together with

$$c_n = \cos k_n, \quad s_n = \sin k_n, \quad \rho^2 = c_1^2 + c_2^2 + c_3^2.$$

For the $+\pi/2$ -flux chirality chosen above,

$$\mathbf{d}(\mathbf{k}) = -2t_f(c_2, -c_3, c_1), \quad d(\mathbf{k}) \equiv |\mathbf{d}(\mathbf{k})| = 2t_f\rho.$$

The Cartesian momentum derivatives are

$$\begin{aligned} \partial_{k_x} \mathbf{d} &= t_f a(s_2, s_3, 2s_1), \\ \partial_{k_y} \mathbf{d} &= \sqrt{3}t_f a(s_2, -s_3, 0). \end{aligned}$$

For compactness, introduce

$$\begin{aligned} A_x &= c_2 s_2 - c_3 s_3 + 2c_1 s_1, & A_y &= c_2 s_2 + c_3 s_3 \\ B_x &= s_2^2 + s_3^2 + 4s_1^2, & B_y &= s_2^2 + s_3^2. \end{aligned}$$

Substitution into the general projector formulas gives the Cartesian quantum metric

$$g_{xx}^- = \frac{a^2}{16} \left(\frac{B_x}{\rho^2} - \frac{A_x^2}{\rho^4} \right), \quad (\text{S211})$$

$$g_{yy}^- = \frac{3a^2}{16} \left(\frac{B_y}{\rho^2} - \frac{A_y^2}{\rho^4} \right), \quad (\text{S212})$$

$$g_{xy}^- = \frac{\sqrt{3}a^2}{16} \left[\frac{s_2^2 - s_3^2}{\rho^2} - \frac{A_x A_y}{\rho^4} \right]. \quad (\text{S213})$$

The corresponding Berry curvature simplifies to

$$\Omega_{xy}^-(\mathbf{k}) = \frac{\sqrt{3}a^2}{4\rho^3} (1 - c_1 c_2 c_3). \quad (\text{S214})$$

Here we used $\mathbf{d} \cdot (\partial_{k_x} \mathbf{d} \times \partial_{k_y} \mathbf{d}) = 4\sqrt{3}t_f^3 a^2 (1 - c_1 c_2 c_3)$. Integrating Eq. (S214) over the magnetic Brillouin zone gives $C_s = +1$ for the occupied band in the chirality convention adopted here. Complex conjugating all hopping amplitudes changes the second component of $\mathbf{d}$ from $-c_3$ to $+c_3$, and hence reverses Ω_{xy}^- and C_s , while leaving d and g_{ij}^- unchanged. The above derivations use the convention $Q_{ij}^- = g_{ij}^- - i\Omega_{ij}^-/2$.

FURTHER ANALYSIS OF RECONSTRUCTION PROTOCOL

Magnitude estimation of physical conductivity

We first discuss the scale of the measured optical signal. The conductivity in our two-dimensional calculation is a genuine sheet conductance, and its dimensionless numerical value is converted to SI units by multiplying by $e^2/\hbar = 2.434 \times 10^{-4}$ S. For the representative parameters used in our numerical calculation, the subgap response is typically of order

$$\begin{aligned} |\text{Im} \sigma_{\text{phys}}^L| &\sim 10^{-4} \frac{e^2}{\hbar} \simeq 2.4 \times 10^{-8} \text{ S}, \\ |\text{Re} \sigma_{\text{phys}}^H| &\sim 10^{-8} \frac{e^2}{\hbar} \simeq 2.4 \times 10^{-12} \text{ S}. \end{aligned}$$

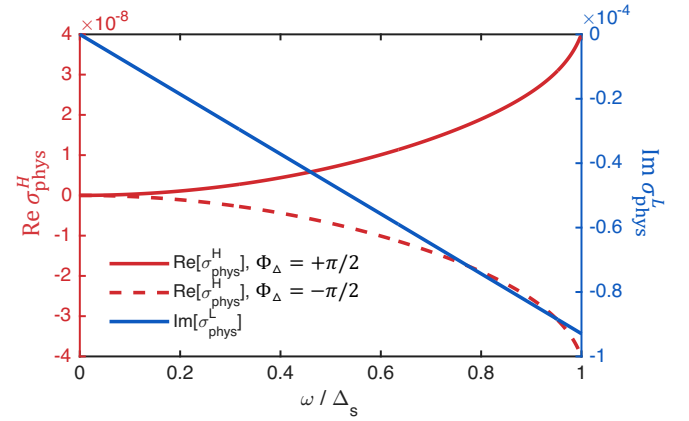


FIG. S5. The 2D longitudinal and Hall conductivities (plotted in units of $e^2/\hbar$) as a function of frequency with $U/t = 15$. Their ratio $[\text{Im} \sigma_{\text{phys}}^L]^2 / \text{Im} \sigma_{\text{phys}}^H$ is proportional to the spinon Chern number. The longitudinal conductivity is invariant under the sign change of the flux, while the Hall conductivity changes sign.

Consequently, in the strict single-layer limit, the Hall sheet conductance is only a few picosiemens. Since the polarization rotation of an optically thin sheet is parametrically of order $Z_0 \sigma_{\text{phys}}^H$, with $Z_0 \simeq 377 \Omega$, the corresponding single-layer rotation is of order 10^{-9} rad or smaller. Such a signal is difficult to resolve with conventional broadband polarimetry, even though the larger longitudinal response may remain accessible to phase-sensitive or resonantly enhanced measurements.

A comparison with experiment must therefore account for the finite thickness of a realistic layered sample. If adjacent layers, separated by a distance d , realize the same CSL chirality, their long-wavelength conductivities contribute to the total response. The equivalent three-dimensional conductivity and the effective sheet conductance of a sample of thickness $L = Nd$ are

$$\sigma_{3D}^{L,H} = \frac{\sigma_{2D}^{L,H}}{d}, \quad G_{\text{eff}}^{L,H} \simeq L \sigma_{3D}^{L,H} = N \sigma_{2D}^{L,H},$$

where the last relation applies in the optically thin, coherently probed limit. A sufficiently thick stack can consequently enhance the absolute Hall signal by many orders of magnitude. Additionally, increasing the number of aligned layers improves the absolute signal-to-noise ratio but does not change the intrinsic hierarchy ratio $|\sigma_{\text{phys}}^H/\sigma_{\text{phys}}^L| \sim 10^{-4}$.

For a concrete estimate, consider an illustrative interlayer spacing $d = 1$ nm. A $1 \mu\text{m}$ -thick sample then contains approximately $N = 10^3$ contributing layers, giving

$$|G_{\text{eff}}^L| \sim 2.4 \times 10^{-5} \text{ S}, \quad |G_{\text{eff}}^H| \sim 2.4 \times 10^{-9} \text{ S}.$$

The Hall response is therefore enhanced from the picosiemens scale to the nanosiemens scale, while $Z_0|G_{\text{eff}}^H| \sim 9 \times 10^{-7}$ corresponds to a polarization rotation of order one micro-radian, up to refractive-index and substrate-dependent factors. For $L = 10 \mu\text{m}$, the same linear estimate gives $|G_{\text{eff}}^H| \sim 2.4 \times 10^{-8} \text{ S}$ and a rotation of order 10^{-5} rad. Thus realistic sample thicknesses can improve the absolute Hall signal by three to four orders of magnitude relative to a single layer.

Finite spinon broadening

We next discuss the low-frequency divergence of the Chern-number estimator with finite spinon broadening, as illustrated in Fig. S6. Here the broadening is implemented by evaluating the full spinon kernels at $z = \omega + i\eta_s$, while the subgap chargin kernel is kept unbroadened. For $\omega, \eta_s \ll \Delta_s$, Eqs. (S127) and (S125) give

$$\Pi_f^L \simeq -\chi_f^L z^2, \quad \Pi_f^H \simeq i\kappa_s z, \quad \Pi_b \simeq -\chi_b \omega^2, \quad \kappa_s \equiv \frac{N_\sigma C_s}{2\pi},$$

where $C_s = \pm 1$ denotes the band Chern number. At sufficiently small ω and η_s , the Hall term dominates the Ioffe-Larkin denominator, $(\Pi_b + \Pi_f^L)^2 + (\Pi_f^H)^2 \simeq -\kappa_s^2 z^2$. The physical kernels consequently reduce to $\Pi_{\text{phys}}^L \simeq -\chi_b \omega^2$ and $\Pi_{\text{phys}}^H \simeq -i\chi_b^2 \omega^4/(\kappa_s z)$. Using $\sigma_{\text{phys}}^{L,H} = i\Pi_{\text{phys}}^{L,H}/\omega$, with $\omega > 0$, we obtain

$$\text{Im} \sigma_{\text{phys}}^L \simeq -\chi_b \omega, \quad \text{Re} \sigma_{\text{phys}}^H \simeq \frac{\chi_b^2}{\kappa_s} \frac{\omega^4}{\omega^2 + \eta_s^2}. \quad (\text{S215})$$

Thus the finite-frequency estimator becomes

$$C_s^{\text{est}}(\omega) \equiv \frac{2\pi}{N_\sigma} \frac{[\text{Im} \sigma_{\text{phys}}^L]^2}{\text{Re} \sigma_{\text{phys}}^H} \simeq C_s \left(1 + \frac{\eta_s^2}{\omega^2}\right). \quad (\text{S216})$$

Finite broadening therefore suppresses the Hall denominator more strongly than the longitudinal numerator, producing a signed divergence as $\omega \rightarrow 0$ at fixed nonzero η_s . The scale $\omega \sim \eta_s$ marks the onset of an appreciable deviation, rather than a pole at $\omega = \eta_s$. The ideal result in main-text Fig. 2 takes $\eta_s = 0$ before the low-frequency limit. For small finite broadening, the appropriate window for estimating the quantized value is instead $\eta_s \ll \omega \ll \Delta_s$.

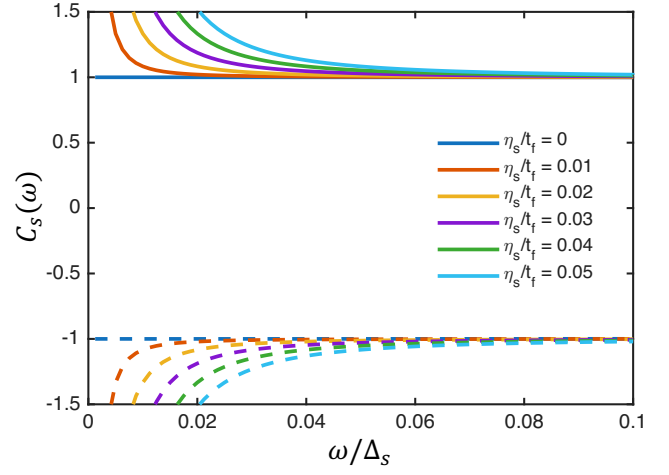


FIG. S6. Extracted spinon Chern number C_s with finite spinon broadening at $U/t = 15$. The dashed lines corresponds to the negative flux $-\pi/2$ result, which yield opposite sign of the Chern number.

Reconstruction error analysis

Although the inverse-response protocol removes the need to model the frequency-dependent chargin kernel, it remains sensitive to experimental errors. According to our mean-field results, we have $|\Pi_b| \ll |\Pi_f^\pm|$. The inverse Ioffe-Larkin rule then gives

$$R_\pm = \frac{1}{\Pi_{\text{phys}}^\pm} = \frac{1}{\Pi_b} + \frac{1}{\Pi_f^\pm} = q_b + r_\pm, \quad |q_b| \gg |r_\pm|. \quad (\text{S217})$$

Thus $\Pi_{\text{phys}}^\pm \simeq \Pi_b$, and the desired spinon information is a small correction to a large common background in $R_\pm$. Let the measured response contains a small error $\delta\Pi_{\text{phys}}^\pm$, so that the measured response is $\tilde{\Pi}_{\text{phys}}^\pm = \Pi_{\text{phys}}^\pm + \delta\Pi_{\text{phys}}^\pm = (1 + \alpha_\pm)\Pi_{\text{phys}}^\pm$, where $|\alpha_\pm| \ll 1$. The real and imaginary parts of $\alpha_\pm$ describe the amplitude and phase errors, respectively. Expanding the reciprocal to first order gives

$$\delta R_\pm \simeq -\frac{\delta\Pi_{\text{phys}}^\pm}{(\Pi_{\text{phys}}^\pm)^2} = -\alpha_\pm(q_b + r_\pm) \quad (\text{S218})$$

Obviously, when $\alpha_\pm$ vanishes, the error $\delta R_\pm$ also vanishes. The relative error in the full inverse response remains of order $|\alpha_\pm|$, however, its size relative to the much smaller spinon term $r_\pm$ is enhanced by $|\Pi_f^\pm/\Pi_b|$. Specifically, we have

$$\frac{|\delta R_\pm|}{|r_\pm|} \simeq |\alpha_\pm| \left| \frac{\Pi_f^\pm}{\Pi_b} \right|. \quad (\text{S219})$$

Whether this large background error survives the extraction depends on the correlation between the two channels.

For the Hall channel, define $\alpha_c = (\alpha_+ + \alpha_-)/2$ and $\alpha_d = (\alpha_+ - \alpha_-)/2$. Using $r_\pm = r_f^L \pm ir_f^H$, the error in their measured

difference is

$$\delta r_f^H = \frac{\delta R_+ - \delta R_-}{2i} \simeq -\alpha_c r_f^H + i\alpha_d(q_b + r_f^L),$$

$$\left| \frac{\delta r_f^H}{r_f^H} \right| \lesssim |\alpha_c| + |\alpha_d| \left| \frac{q_b}{r_f^H} \right|. \quad (\text{S220})$$

The last estimate uses $|q_b| \gg |r_f^L|$ and assumes $r_f^H \neq 0$. An identical calibration error α_c in both $\pm$ channels cancels the large background and primarily rescales the Hall response. A small relative error α_d , however, leaves a residual proportional to q_b , which can exceed the spinon signal. Accurate Hall extraction therefore requires $|\alpha_d| \ll |r_f^H/q_b|$.

The longitudinal absorptive channel is similarly sensitive to phase calibration. For example, there could be a small phase error in the measured response $\tilde{\Pi}_{\text{phys}}^\pm = e^{i\phi_\pm} \delta \Pi_{\text{phys}}^\pm \simeq (1 + \alpha_\pm) \Pi_{\text{phys}}^\pm$, or equivalently, $\alpha_\pm = i\phi_\pm$. In this case, the relative phase error $\alpha_+ = -\alpha_-$ does not affect the longitudinal response, but an identical phase error $\alpha_+ = +\alpha_- = i\phi$, with real ϕ , gives

$$\delta[\text{Im} r_f^L] = \text{Im} \frac{\delta R_+ + \delta R_-}{2} \simeq -\phi \text{Re}(q_b + r_f^L) \simeq -\phi q_b. \quad (\text{S221})$$

Here q_b is real below the chargon absorption threshold. For an accurate reconstruction of the spinon longitudinal response, one may require the common phase error to satisfy $|\phi| \ll |\text{Im} r_f^L/q_b|$. These estimates explain why a small chargon kernel may impose stringent error control in the amplitude and phase calibration measurements.

Another possible error source is the calibration of the spinon interband threshold Δ_s , which fixes the KK subtraction constant through $r_{f,\infty}^L \simeq 2.49788/\Delta_s$. If the inferred threshold is deviated from the true value by a small relative error as $\tilde{\Delta}_s = (1 + \varepsilon)\Delta_s$, then

$$\tilde{r}_{f,\infty}^L = \frac{r_{f,\infty}^L}{1 + \varepsilon}, \quad \frac{\delta r_{f,\infty}^L}{r_{f,\infty}^L} = -\frac{\varepsilon}{1 + \varepsilon} \simeq -\varepsilon. \quad (\text{S222})$$

With the optical input held fixed, this produces a frequency-independent shift in the reconstructed $\text{Re} r_f^L$. However, the final inversion $\Pi_f^\pm = 1/(r_f^L \pm i r_f^H)$ converts this offset into a frequency-dependent error, especially where the inverse spinon response is small. The reconstructed geometric spectra can therefore undergo changes in line shape rather than a simple overall rescaling.

Figure S7 illustrates this sensitivity for threshold errors of 1%, 3%, and 5%, with the underlying response otherwise unchanged. The discrepancy grows with the calibration error; for the parameters shown, a 5% error produces a large deviation from the exact result for both the quantum-metric and Berry-curvature spectra, including reduced peak heights and enhanced high-frequency shoulders. We admit that our reconstruction protocol of spinon QGT spectral is feasible in principle, but quantitatively accurate experimental reconstruction demands precise measurements of the complex optical

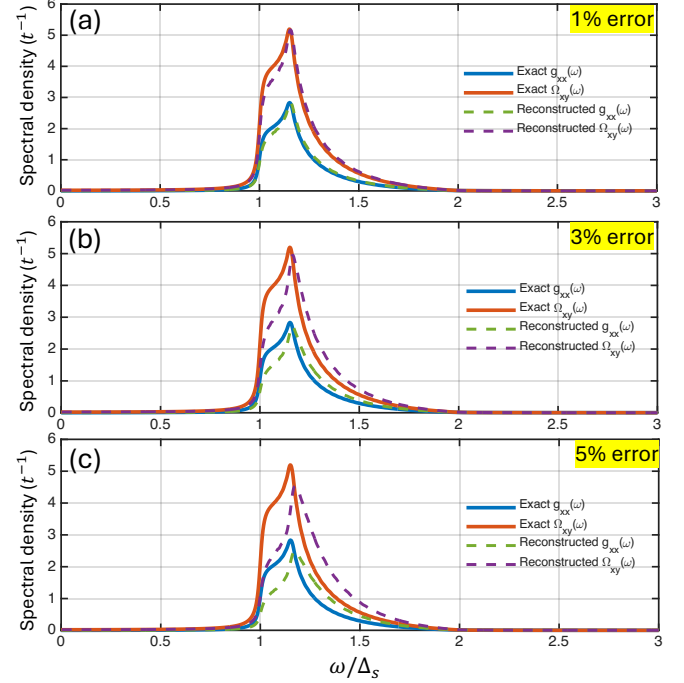


FIG. S7. Spectral density of Berry curvature and quantum metric when the spinon gap Δ_s is measured with 1% error (a), 3% error (b) and 5% error (c). These errors are simulated by manually deviating the spinon gap $\Delta_s = 2\sqrt{3}t_f$ in the reconstruction procedure. The dashed lines corresponds to reconstructed results while the invariant solid lines corresponds to the exact results. The calculation is performed at $U/t = 15$ with positive flux $\pi/2$ and $\eta_s/t_f = 0.05$.

response and Δ_s , especially in the deep-mott regime. We expect that a smaller Hubbard interaction U (but still in the CSL phase) would reduce the precision requirement for the reconstruction when the virtual doublon-holon excitations become less forbidden, yielding a larger chargon response.